

Comparative Research Analysis: Behavior of Null and Timelike Geodesics around Black Holes, Naked Singularities, and Wormholes

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DECLARATION

I hereby declare that the dissertation “**Behavior of Null and Timelike Geodesics around Black Holes, Naked Singularities, and Wormholes**” submitted by me to Charotar University of Science and Technology in partial fulfilment of the requirements of the degree of Master of Science Physics is the record of work carried out by me during the period from January, 2022 to April, 2022 under the supervision of *Dr. Dipanjan Dey*. I also declare that this work has not formed the basis for the award of any degree, diploma or titles by any other Institution or University.

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CERTIFICATE

This is to certify that the dissertation entitled ” *Comparative study of behavior of Null and Timelike Geodesics around Black Holes, Naked Singularities, and Wormholes*” submitted by *Kauntey Rajeshbhai Acharya* (ID No. 20PHY001), to the Charotar University of Science and Technology in partial fulfillment of the requirements of the degree of Master of Science (Physics) is a bonafide record of research project work carried out by him under our supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for award of any degree, diploma, or titles.

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ABSTRACT

The solutions to Einstein's field equations allow for the existence of a wide variety of compact objects, each arising from different spacetime metrics. These metrics determine the physical properties and geometric structure of the objects, which in turn influence how they interact with matter and radiation in their surroundings. Because many compact objects can appear similar in mass or gravitational strength, it becomes essential to study observables such as their shadows and orbital dynamics. These features provide crucial information that helps distinguish one type of compact object from another.

In this thesis, I explore several key aspects of these observables, focusing on the shadow structure and the behaviour of timelike orbits associated with black holes, naked singularities, and wormhole spacetimes. Understanding how particles and photons move in these geometries sheds light on the fundamental differences between them, especially in strong-gravity regimes where classical intuition fails.

In particular, I analyse both null geodesics and timelike geodesics for a specific traversable wormhole model proposed in the literature. By examining how light rays and massive particles propagate in this spacetime, we gain insight into the characteristic signatures that may help identify such exotic objects through astrophysical observations.

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Chapter 1

Introduction

As *Homo sapiens*, we have lived on Earth, a planet orbiting the Sun, which is only one among billions of stars moving through the many galaxies of the universe. Throughout our history, we have sought to understand the cosmos and questioned the forces that shape it. A central inquiry has been: What causes celestial bodies to move around one another? Numerous explanations were proposed, but the first scientific account came from Isaac Newton through his Law of Gravitation. According to this law, every particle of matter attracts every other particle with a force proportional to the product of their masses and inversely proportional to the square of the distance between them.

Newton's formulation clarified why planets orbit the Sun and why the Moon orbits Earth. It revealed that everyday phenomena such as falling objects, planetary motion, and ocean tides arise from the same fundamental principle. This insight transformed science and technology, making possible achievements such as satellite launches and lunar missions. For the most part, the universe appeared to follow this law, with only a few notable exceptions.

Albert Einstein's Special Theory of Relativity (1905) and General Theory of Relativity (1915) fundamentally transformed our understanding of gravity. The Special Theory of Relativity is built upon two postulates proposed by Einstein. The first

states that the laws of physics remain the same in all inertial frames of reference. The second states that light travels through a vacuum at a constant speed c , which is the same for every inertial observer, independent of the motion of the light source. From these principles, Einstein was able to explain several relativistic phenomena, including length contraction, time dilation, and the increase of mass with velocity.

Einstein's General Theory of Relativity introduced the idea that gravity is not a traditional force but rather a manifestation of the curvature of spacetime. In doing so, it addressed the long-standing question left open by Newton's theory regarding how two bodies exert gravitational influence on each other. According to Einstein, every massive object deforms the spacetime surrounding it in proportion to its mass, and nearby bodies move along paths determined by this curved geometry. John Wheeler summarized this elegantly: "Matter tells spacetime how to curve, and curved spacetime tells matter how to move." The theory successfully explained phenomena that Newtonian gravity could not, such as the anomalous precession of Mercury's orbit. Further developments in this framework also led to theoretical descriptions of exotic astrophysical objects, including black holes, neutron stars, and quasars.

The General Theory of Relativity also provides a framework for understanding the evolution of stars and their gravitational collapse into various possible end states. The earliest model of such collapse for a spherically symmetric, homogeneous dust cloud was developed by Oppenheimer, Snyder, and Datt. Their model predicts that the collapse results in a black hole. In the Oppenheimer–Snyder–Datt framework, trapped surfaces appear before the formation of the central spacelike singularity. As a result, the singularity becomes causally disconnected from the rest of spacetime, which indicates the presence of an event horizon. However, the model is highly idealized because it assumes uniform density and zero pressure inside the collapsing star. Subsequent research introduced more realistic conditions, such as inhomoge-

neous matter distributions and non-zero pressure profiles. Studies that consider a higher central density, which is physically more reasonable, show that the collapse can lead to a strong curvature singularity that may be visible either locally or globally, depending on the equilibrium conditions.

Both black holes and naked singularities arise as possible end states of gravitational collapse and are solutions to Einstein's field equations. There exists another category of compact objects, known as wormholes, which also satisfy these equations. Wormholes are structurally distinct from black holes and naked singularities because they do not contain a central spacetime singularity. Instead, they possess a throat region connecting two separate regions of spacetime. Although there is no accepted physical mechanism for their formation, wormholes remain theoretically appealing and are often considered when exploring the types of compact objects that may exist at galactic centers.

Distinct geometric features of these compact objects lead to differences in the observable phenomena around them. One important example is the bending of light due to strong gravitational fields. Light emitted by distant sources can be deflected or captured, creating a low-intensity central region known as the shadow of the compact object. This shadow is determined by the behaviour of null geodesics, which describe the paths of light rays. The motion of massive particles, described by timelike geodesics, can also be examined. By studying both null and timelike geodesics, one can gain insight into the underlying spacetime geometry of compact objects. Beyond shadow formation, gravitational lensing provides further information by analysing the bending of light rays, the time delays experienced by particles near the object, and the associated deflection angles in both strong and weak regimes. Since these effects depend directly on spacetime geometry, they serve as valuable tools for distinguishing between different compact objects and for probing the nature of those located at galactic centers.

In this thesis, we explore timelike and null geodesics, as well as shadows, of dense structures like black holes, naked singularities, and wormholes using concepts from the General Theory of Relativity. We also address gravitational lensing and weak deflection angles later in the thesis. The thesis is organized as follows: Chapter Two introduces basic concepts of the General Theory of Relativity. Chapter Three discusses the Schwarzschild and Kerr black holes, focusing on their shadow shapes. In Chapter Four, we first examine the formation of naked singularities from gravitational collapse, followed by the JMN-1 and JNW naked singularities, and then study their shadows. Chapter Five covers the basic geometrical and physical properties of wormholes, including the shadow of Morris-Thorne wormholes. Chapter Six analyzes a wormhole model from the literature and the behavior of null and timelike geodesics around it. In Chapter Seven, we briefly discuss gravitational lensing before concluding with results and discussions.

Chapter 2

General Relativity

2.1 Equivalence Principle

The general theory of relativity is built upon the fundamental idea known as the equivalence principle. Its earliest formulation, often called the weak equivalence principle or the Galilean principle of the universality of free fall, states that an object's inertial mass and gravitational mass are identical. As a result, the gravitational force experienced while standing on a massive body is locally indistinguishable from the pseudo-force felt by an observer inside an accelerating frame of reference. Einstein extended this idea and formulated what is now known as the Einstein equivalence principle. According to this principle, within sufficiently small regions of spacetime, the laws of physics take the same form as those of special relativity, and no experiment can determine whether a gravitational field is present. The equivalence principle therefore implies that gravity arises from the geometry of spacetime rather than from a traditional force.

Since gravity is understood as a manifestation of spacetime geometry, it becomes essential to describe this geometry mathematically. This leads to the introduction of the metric tensor, which serves as the fundamental tool for measuring distances, intervals, and the curvature of spacetime within the framework of general relativity.

2.2 Metric Tensor

The metric tensor $g_{\mu\nu}$ is the central mathematical object used in general relativity to describe the geometry of spacetime. It defines how distances, intervals, and angles are measured, and determines how spacetime curvature influences the motion of particles and light. Every physical prediction in general relativity ultimately depends on the form of the metric, making it the foundation for studying different gravitational systems.

Information about spacetime geometry is obtained from the metric tensor $g_{\mu\nu}$. The spacetime interval is defined as,

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (2.1)$$

where summation over the indices μ and ν is implied. The indices take the values 0, 1, 2, 3, corresponding to the coordinates t, x, y , and z . The components of the metric tensor specify how each coordinate contributes to the line element ds^2 , and from these components one can determine the curvature of spacetime. Equation (1) may be written in the form

$$ds^2 = -g_{tt}dt^2 + g_{rr}dr^2 + g_{\theta\theta}d\theta^2 + g_{\phi\phi}d\phi^2 \quad (2.2)$$

The full metric tensor, containing all possible components, can be expressed in the general form

$$g_{\mu\nu} = \begin{pmatrix} g_{tt} & g_{tr} & g_{t\theta} & g_{t\phi} \\ g_{rt} & g_{rr} & g_{r\theta} & g_{r\phi} \\ g_{\theta t} & g_{\theta r} & g_{\theta\theta} & g_{\theta\phi} \\ g_{\phi t} & g_{\phi r} & g_{\phi\theta} & g_{\phi\phi} \end{pmatrix} \quad (2.3)$$

however, in the equation, (2), non-diagonal components are zero. The matrix shows how one can multiply differentials $dt, dr, d\theta, d\phi$ which we can write as a table

$$\begin{aligned}
& dt \quad dr \quad d\theta \quad d\phi \\
& dt \quad g_{tt} \quad g_{tr} \quad g_{t\theta} \quad g_{t\phi} \\
[g_{\mu\nu}] = & dr \quad g_{rt} \quad g_{rr} \quad g_{r\theta} \quad g_{r\phi} \\
& d\theta \quad g_{\theta t} \quad g_{\theta r} \quad g_{\theta\theta} \quad g_{\theta\phi} \\
& d\phi \quad g_{\phi t} \quad g_{\phi r} \quad g_{\phi\theta} \quad g_{\phi\phi}
\end{aligned} \tag{2.4}$$

The components of the metric tensor $g_{\mu\nu}$ therefore encapsulate the curvature of spacetime and serve as the basis for analysing geodesics, gravitational effects, and the overall structure of any spacetime model in general relativity. However, In the special case where spacetime is completely flat and exhibits no curvature, the metric tensor takes the form,

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{2.5}$$

The corresponding spacetime interval is then expressed as,

$$ds^2 = -c^2 dt^2 + dr^2 + d\theta^2 + d\phi^2 \tag{2.6}$$

or, in Einstein summation notation,

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu \tag{2.7}$$

The matrix above is known as the Minkowski metric, and the associated line element represents a flat spacetime with no curvature.

The sign of the spacetime interval between two nearby events determines whether they are causally related. If $ds^2 < 0$, the separation is timelike, meaning the events

can be connected by the worldline of a material particle. If $ds^2 = 0$, the separation is null, and the events lie along the worldline of a photon. If $ds^2 > 0$, the separation is spacelike, and no massive particle or photon can travel between the events. Time-like and null intervals correspond to causally connected events, whereas spacelike intervals indicate that the events are causally disconnected.

Once the metric tensor is specified, it provides complete information about the geometry of spacetime. From this geometric structure, one can determine how massive particles and photons move through spacetime. To describe these trajectories rigorously, the geodesic equation must be introduced.

2.3 Geodesic

A geodesic represents the path of shortest distance between two points in a given spacetime. In flat space, the geodesic connecting two points is simply the straight line between them. On a spherical surface, however, the geodesic is the arc of a great circle that joins the two points. To extend this concept to curved spacetime, where no simple geometric shape can be assumed, one requires a mathematical tool known as the Christoffel symbol, or connection coefficient, which provides the means to define geodesics in a general curved geometry.

2.3.1 Christoffel Symbol or Connection Coefficient

In general relativity, the study of spacetime geometry requires understanding how the components of the metric tensor vary with respect to spatial and temporal coordinates. However, when differentiating the metric tensor, it is also necessary to account for how the basis vectors change with position in spacetime. This variation of the basis vectors is captured by the Christoffel symbols, also known as connection coefficients.

Consider a vector $V = V^\alpha e_\alpha$, where V^α represents the components of the vector and e_α denotes the basis vectors. To examine how V changes with respect to the

coordinate to x^β , we take the derivative:

$$\frac{dV}{dx^\beta} = \frac{dV^\alpha}{dx^\beta} e_\alpha + V^\alpha \frac{de_\alpha}{dx^\beta} \quad (2.8)$$

The Christoffel symbols are introduced through the relation

$$\Gamma_{\alpha\beta}^\lambda e_\lambda = \frac{de_\alpha}{dx^\beta} \quad (2.9)$$

which shows that $\Gamma_{\alpha\beta}^\lambda$ quantifies the rate at which the basis vector e_α changes with respect to x^β in the direction of the basis vector e^λ . These symbols are called connection coefficients because they describe how neighbouring points on a curved manifold are connected through changes in the basis vectors.

The Christoffel symbols can be expressed in terms of the metric tensor as

$$\Gamma_{\alpha\beta}^\lambda = \frac{1}{2} g^{\lambda\mu} \left(\frac{dg_{\mu\beta}}{dx^\alpha} + \frac{dg_{\alpha\mu}}{dx^\beta} - \frac{dg_{\alpha\beta}}{dx^\mu} \right) \quad (2.10)$$

This expression allows us to compute the Christoffel symbols directly from the metric tensor, providing insight into how the geometry of spacetime changes from point to point. With the Christoffel symbols defined, we are equipped to formulate the geodesic equation, which describes the motion of particles and photons in any curved spacetime.

2.3.2 Null Geodesic

To understand how light rays propagate through curved spacetime, we use the geodesic equation constructed from the Christoffel symbols. Since light travels along paths determined entirely by the geometry of spacetime, its trajectory is described by a null geodesic. This geodesic equation is written as,

$$\frac{DU_\alpha}{d\lambda} = \frac{d^2 x^\alpha}{d\lambda^2} + \Gamma_{\gamma\beta}^\alpha \frac{dx^\beta}{d\lambda} \frac{dx^\gamma}{d\lambda} \quad (2.11)$$

where λ is an affine parameter that parametrizes the curve in a consistent way.

For the motion of massive particles, the relevant trajectories are timelike geodesics. In this case, the proper time τ can be used instead of the affine parameter, giving the expression

$$\frac{DU_\alpha}{d\tau} = \frac{d^2x^\alpha}{d\tau^2} + \Gamma_{\gamma\beta}^\alpha \frac{dx^\beta}{d\tau} \frac{dx^\gamma}{d\tau} \quad (2.12)$$

These geodesic equations allow us to determine the paths followed by photons and massive particles as they move through spacetime. Their motion is governed solely by the curvature encoded in the metric and occurs without the action of any external forces; the gravitational field is entirely geometric.

To describe this gravitational field more fully, we require a source term that captures how matter and energy are distributed throughout spacetime. This role is played by the stress-energy tensor, which specifies the density and flow of energy and momentum at every event in spacetime.

2.4 Stress Energy Tensor

The stress-energy tensor $T^{\mu\nu}$ provides information about the motion and distribution of gravitating matter throughout spacetime. Through this tensor, one can determine how matter and energy generate the gravitational field.

In many physical situations, the stress-energy tensor is modeled as that of a perfect fluid. In such a description, the material content of the system is treated as a smooth, continuous medium. Each fluid element is assumed to behave uniformly within a small neighbourhood, allowing us to describe its dynamics using averaged physical quantities such as density, pressure, and temperature defined at that local point.

2.4.1 Dust

A simple fluid element is one in which the constituent particles do not interact internally. In the case of dust, all particles are at rest relative to a locally defined comoving frame. At any given moment in this frame, the particles have no random motion within the dust element.

In a real fluid, however, particles exhibit random motion relative to one another, which gives rise to pressure. Additionally, the particles interact through various forces. If a relativistic fluid has no internal interactions such as viscosity or heat conduction, it is classified as a perfect fluid. For dust, there are no interactions between particles, and because all particles have zero velocity with respect to the comoving frame, the fluid element is pressureless.

Using the stress-energy tensor $T^{\mu\nu}$, the change in the components of four-momentum Δp^α across a hypersurface can be expressed as

$$\Delta p^\beta = T^{\alpha\beta} \Delta S_\alpha \quad (2.13)$$

In the case of dust, since there is no relative motion among the particles, the change in momentum is zero. This implies that $T^{ij} = 0$ for $i, j = 1, 2, 3$. The component T^{00} , however, remains non-zero and represents the energy density of the dust element.

2.4.2 Perfect Fluid

When we treat a relativistic fluid as a perfect fluid, its particles possess non-zero velocities, which gives rise to pressure within the fluid element. However, a perfect fluid is defined by the absence of internal interactions such as viscosity or heat conduction. Because there is no bulk motion relative to the comoving frame and no heat transfer between particles, the mixed components of the stress-energy tensor satisfy $T^{i0} = 0$ for $i = 1, 2, 3$, indicating that the energy flux vanishes.

For the spatial off-diagonal components, $T^{0i} = 0$ for $i = 1, 2, 3$ is zero because a perfect fluid has no shear or viscous stresses. The diagonal spatial components $T^{ii} = 0$ represent the force per unit area, corresponding to the flux of the i^{th} momentum component across a surface normal to the x^i direction. These components are identified with the pressure P .

Thus, the stress-energy tensor for a perfect fluid takes the matrix form,

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix} \quad (2.14)$$

where ρ denotes the energy density.

From this tensor, one can determine both the distribution and motion of matter that produces gravitational effects. Up to this point, we have discussed how gravitating matter influences spacetime, but we have not yet introduced the mathematical concept used to quantify spacetime curvature. This curvature is described using the Riemann curvature tensor.

2.5 Riemann Curvature Tensor

The Riemann curvature tensor provides the mathematical description of spacetime curvature in general relativity. It is constructed from the Christoffel symbols, or connection coefficients, which themselves are derived from the spacetime metric and therefore contain information about the geometric structure of spacetime. The Riemann curvature tensor expressed in terms of Christoffel symbols is

$$R^\alpha_{\beta\mu\nu} = \frac{d\Gamma^\alpha_{\beta\nu}}{dx^\mu} - \frac{d\Gamma^\alpha_{\beta\mu}}{dx^\nu} + \Gamma^\rho_{\beta\nu}\Gamma^\sigma_{\rho\mu} - \Gamma^\rho_{\beta\mu}\Gamma^\sigma_{\rho\nu} \quad (2.15)$$

For flat spacetime, all components of the Riemann tensor vanish, so $R^\alpha_{\beta\mu\nu} = 0$.

At this stage, we have mathematical tools that describe spacetime curvature and, separately, the distribution and motion of matter through the stress-energy tensor. To understand how matter shapes the geometry of spacetime and how that geometry governs the motion of matter and radiation, we seek an equation that links these two mathematical quantities.

The Riemann tensor has four indices, while the stress-energy tensor has two. To relate them, we first contract the Riemann tensor to obtain the Ricci tensor:

$$R_{\alpha\nu} = g^{\beta\mu} R_{\beta\mu\nu}^{\alpha} \quad (2.16)$$

A further contraction using the inverse metric gives the Ricci scalar:

$$R = g^{\beta\nu} R_{\beta\nu} \quad (2.17)$$

The Ricci tensor can also be written in contravariant form by raising indices using the metric tensor:

$$R^{\rho\sigma} = g^{\rho\beta} g^{\sigma\nu} R_{\beta\nu} \quad (2.18)$$

With these quantities, one can define the Einstein tensor:

$$G_{\rho\sigma} = R_{\rho\sigma} - \frac{1}{2} g_{\rho\sigma} R = \kappa T_{\rho\sigma} \quad (2.19)$$

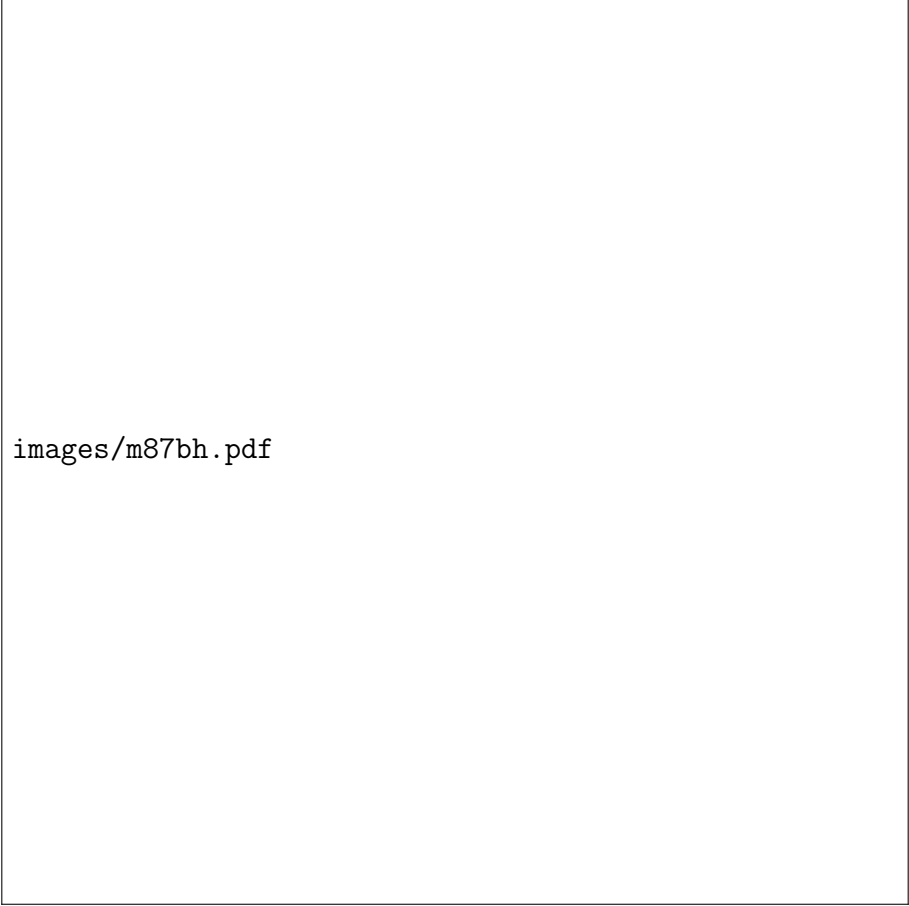
which is the Einstein field equation. This equation forms the core of general relativity and relates the curvature of spacetime to the energy and momentum of matter. By solving the Einstein field equations under different assumptions, one obtains various spacetime geometries corresponding to physical situations such as black holes, expanding universes, stellar interiors, and more. In the next chapter, we discuss black holes.

Chapter 3

Black Holes and Physical Properties

Stars form from large concentrations of dust and gas known as molecular clouds. These clouds collapse under their own gravity and give rise to protostars. Based on their initial mass, stars are grouped into three categories: low-mass stars, intermediate-mass stars, and massive stars. All types of stars produce energy through nuclear fusion. At the beginning of their life cycle, hydrogen nuclei fuse to form helium, releasing a large amount of energy in an exothermic process that counteracts the inward pull of gravity. After the hydrogen in the core is exhausted, fusion continues with heavier elements such as carbon, oxygen, neon, silicon, and iron. Stars that undergo these fusion stages are known as main sequence stars.

Fusion cannot proceed beyond iron because heavier elements require energy to fuse. The thermal pressure generated by fusion eventually becomes insufficient to balance gravity, and the star begins to collapse under its own weight. Low-mass and intermediate-mass stars reach stable configurations through quantum mechanical pressures. Electron degeneracy pressure supports white dwarfs, and neutron degeneracy pressure supports neutron stars. However, more massive stars cannot halt the gravitational collapse through any known form of quantum pressure. According to general relativity, this continued gravitational collapse results in a



images/m87bh.pdf

Figure 3.1: Black hole at the center of M87 Galaxy

spacetime singularity in which matter density and gravitational forces grow without bound [**penrose1965gravitational**]. This extremely dense region can be studied by examining different possible astrophysical compact objects such as black holes, naked singularities, boson stars, and similar configurations [**joshi1993naked**, **visinelli2021boson**, **janis1968reality**, **joshi2011equilibrium**]. These compact objects exhibit interesting physical and geometric properties, and distinguishing among them is an important issue in observational astrophysics.

3.1 Black hole

The term "Black hole" is used for a compact astrophysical object from which nothing can escape to a faraway observer. Even light cannot get away from its trapped surface (an event horizon) due to extreme gravitational attraction. However, a dis-

inct signature of a black hole is an "Event Horizon" that indicates the trapped surface of a black hole. Within the general theory of relativity, the spacetime singularity can form at the center of such a black hole which is the actual boundary of the spacetime manifold [**penrose1965gravitational**]. The spacetime singularity is the caustic point where the geodesics are incomplete, and all the physical quantities such as matter densities, pressures, and spacetime curvatures are infinite. In very broad way, when the singularity is covered by an event horizon, it is known as black hole.

In the general theory of relativity, there are mainly two well-known non-rotating black holes, (1) the Schwarzschild black hole and (2) the Reissner-Nordstrom black hole. The Schwarzschild black hole is a vacuum solution of the Einstein equations, which has no charge and no rotation. Whereas, the Reissner Nordstrom's black hole is a static solution of the Einstein-Maxwell field equations with electric charge without rotation. There also exist rotating generalizations of these static black holes are given as the Kerr (with zero charge) and Kerr-Newman (with non-zero charge) black holes respectively [**hartle2003gravity**]. In this thesis, we discuss only charge-less Schwarzschild and Kerr black holes.

3.2 Schwarzschild Black Hole

In 1916, Karl Schwarzschild found the first exact vacuum solution to the Einstein's field equations [**schwarzschild2003gravitational**]. The Schwarzschild black hole spacetime in spherical coordinates can be written as,

$$ds_{SCH}^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad (3.1)$$

where $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$ represents the metric of the two spheres, and M is the total mass of the black hole. It can be seen that, there exists a central spacetime singularity occurs at $r = 0$. At $r = 2M$, there is a presence of event horizon which is basically a null hypersurface. The radius of the event horizon is also known as

the Schwarzschild radius. The central spacetime singularity at $r = 0$ is covered by the event horizon at $r = 2M$. The Schwarzschild metric is a vacuum solution to Einstein's field equations. The energy density and pressures of this spacetime are,

$$\rho = 0, \quad p_r = p_\theta = p_\phi = 0. \quad (3.2)$$

where, ρ is the energy density of the matter, while p_r , p_θ and p_ϕ are pressures components in corresponding directions respectively. The weak, strong and null energy conditions of this spacetime can be easily verified by the following conditions,

$$\rho \geq 0, \quad \rho + p_r \geq 0, \quad \rho + p_\theta \geq 0, \quad \rho + p_\phi \geq 0, \quad \rho + p_r + 2p_\theta \geq 0. \quad (3.3)$$

This black hole spacetime considers non rotating, spherically symmetric and non charged geometry.

3.3 Shadow

For any far away observer, the shadow consists of dark region in the center of the structure of the compact object and bright region with radially decreasing brightness [joshi2020shadow]. Shadow for a black hole at the center of the M87 galaxy can be seen in 3.1.

Lightrays coming from any source in the sky and passing from a region nearby any compact object would cast a shadow. The central dark region exists as the light rays very close to the compact object would plunge inside the object because of the highly-dense structure and intense gravitational pull from the compact object. However not all the light rays plunge into the compact object. Some light rays just deflect and the light rays very far from the region of the high gravitational pull of the compact object go undeflected. So to decide at which distance the rays will start deflecting and which rays will get absorbed, one needs to find out the effective

potential energy of the system.

The study of null-like geodesics is useful to describe the formation of the shadows cast by ultra-compact objects. In this chapter, we study the shadow cast by Schwarzschild black hole and Kerr black hole.

3.4 Null-like geodesic in spherically symmetric and static spacetime

Light rays can be mathematically described using null geodesics. To describe the world line of a light ray, we consider a coordinate x^α as a function of an affine parameter λ . Here the null vector $k^\alpha = dx^\alpha/d\lambda$ is tangent to the world line. The general spherical symmetric and static spacetime can be written as,

$$ds^2 = -g_{tt}(r)dt^2 + g_{rr}(r)dr^2 + g_{\theta\theta}(r)d\theta^2 + g_{\phi\phi}(r)\sin^2\theta d\phi^2, \quad (3.4)$$

where g_{tt} , g_{rr} , $g_{\theta\theta}$ and $g_{\phi\phi}$ are functions of radial coordinate r . For this general spacetime we conserved quantities of energy and momentum corresponding to null geodesics can be defined as,

$$\gamma = -\xi \cdot k = g_{tt}(r) \left(\frac{dt}{d\lambda} \right), \quad (3.5)$$

$$h = \eta \cdot k = g_{\phi\phi}(r) \left(\frac{d\phi}{d\lambda} \right), \quad (3.6)$$

where λ is an affine parameter, ξ and η are killing vectors corresponding to symmetries. Here, γ and h represent the conserved energy and angular momentum per photon. The normalization of null vectors for null-like geodesics is defined as,

$$k \cdot k = g_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0, \quad (3.7)$$

Now, using the conserved quantities and normalization condition for the above general spherically symmetric and static spacetime (3.4) for null-like geodesic we get the following expression,

$$-\frac{\gamma^2}{g_{tt}(r)} + g_{rr}(r) \left(\frac{dr}{d\lambda} \right)^2 + \frac{h^2}{g_{\phi\phi}(r)} = 0. \quad (3.8)$$

Here it is considered that null geodesics are moving in an equatorial plane $\theta = \pi/2$ for simplicity. By multiplying $g_{tt}(r)/h^2$ term to the above equation we get,

$$\frac{1}{b^2} = \frac{g_{tt}(r)g_{rr}(r)}{h^2} \left(\frac{dr}{d\lambda} \right)^2 + V_{eff}(r), \quad (3.9)$$

where, $b = h/\gamma$ is the impact parameter and $V_{eff}(r)$ is the effective potential for photon orbits expressed as,

$$V_{eff}(r) = \frac{g_{tt}(r)}{g_{\phi\phi}(r)}. \quad (3.10)$$

The phenomena of plunging or deflection of null geodesics can be defined using the above expression of effective potential. We can get the unstable circular null-like orbits when the following conditions are satisfied,

$$V'_{eff}(r_{ph}) = 0, \quad V''_{eff}(r_{ph}) < 0, \quad (3.11)$$

Here r_{ph} is the radius of the photon sphere. The above conditions imply that effective potential has a maximum value at $r = r_{ph}$ for unstable circular photon orbits. The geometrical shape of the spacetime is defined by the metric components. Since null geodesics follow the curvature of the spacetime, it can affect the trajectories of light rays. Using the equations (3.6) and (3.9), the expression describing the shape of orbits of light ray can be defined as,

$$\left(\frac{dr}{d\phi} \right)^2 = \pm \frac{g_{\phi\phi}^2(r)}{g_{tt}(r)g_{rr}(r)} \left[\frac{1}{b^2} - V_{eff}(r) \right]. \quad (3.12)$$

Here, the $\pm$ sign represent the outgoing and incoming light ray path respectively. By differentiating above eq. (3.12) with respect to ϕ and considering $u = \frac{1}{r}$, the equation of orbits of null geodesics can be written as,

$$\frac{d^2 u}{d\phi^2} + \frac{g_{\phi\phi}^2(u)u^4}{2g_{tt}^2(u)g_{rr}(u)b^2} \left(\frac{dg_{tt}(u)}{du} \right) - A(u) \left(\frac{dg_{rr}(u)}{du} \right) + B(u) \left(\frac{dg_{\phi\phi}(u)}{du} \right) - C(u) = 0 , \quad (3.13)$$

and,

$$A(u) = \left[\frac{g_{\phi\phi}^2(u)u^4}{2g_{tt}(u)g_{rr}^2(u)b^2} - \frac{g_{\phi\phi}(u)u^4}{2g_{rr}^2(u)} \right] , \quad (3.14)$$

$$B(u) = \left[\frac{u^4}{2g_{rr}(u)} - \frac{g_{\phi\phi}(u)u^4}{g_{tt}(u)g_{rr}(u)b^2} \right] , \quad (3.15)$$

and

$$C(u) = \left[\frac{2g_{\phi\phi}^2(u)u^3}{g_{tt}(u)g_{rr}(u)b^2} - \frac{2g_{\phi\phi}(u)u^3}{g_{rr}(u)} \right] . \quad (3.16)$$

The above equation (3.13) represents the light ray orbits for a general spherically symmetric and static spacetime which can be solved numerically for a given space-time.

3.5 Observable intensity

It is widely studied that the shadows of compact objects can be formed due to a photon sphere. The photon sphere is the sphere corresponding to the unstable circular light ray orbits. In addition to the existence of a photon sphere, one more condition has to be satisfied, and that is the effective potential of the spacetime for the null-like geodesics should have an upper bound at $r = r_{ph}$. The behavior of the light rays near the compact object can be understood by the condition of impact parameter b . From the equation (3.9), we can write the impact parameter corresponding to the turning point of the light ray,

$$b_{tp} = \frac{r_{tp}}{\sqrt{g_{tt}(r_{tp})}} . \quad (3.17)$$

where $dr/d\lambda = 0$ at the turning point and r_{tp} is the radius of the turning point from the compact object. Note that, for null geodesics, if only one extremum value of effective potential exists and that extremum value is the unstable photon orbit or maximum value of potential, then the minimum impact parameter at the turning point of null geodesics is,

$$b_{tp} = b_{ph}, \quad (3.18)$$

where b_{ph} is the impact parameter corresponding to the photon sphere. The light rays coming from the background source with the impact parameter $b < b_{ph}$ could not reach an asymptotic observer, they would be trapped inside the photon sphere. Hence, the shadow with radius b_{ph} would form in the observer's sky. There are numbers of literature where authors show that the shadow can form without a photon sphere in particular types of geometries [**dey2021shadow**]. However, from that literature, we noticed that if the photon sphere is absent in a given spacetime and the effective potential diverges near the center, then the shadow will not occur in that spacetime [**shaikh2022shadows**].

Now, it is crucial to visualize the shadow cast by any compact object in a physical situation. We need to calculate the intensity distribution of the light rays coming from the accretion matter around the compact object. As we know, the radially freely falling thin accretion matter radiates monochromatic light, where the emission rate of that light per unit volume describe as,

$$j(\nu_e) \propto \frac{\delta(\nu_e - \nu_*)}{r^2}, \quad (3.19)$$

where ν_e is the emitted frequency of a photon as measured in the rest frame of the emitter. Note that, for simplicity, in this section, we have considered the case in which the thin accretion matter is in spherically symmetric form. The observed intensity of the light coming from the accretion matter could be obtained in the

asymptotic observer's sky with the coordinates X, Y as,

$$I_{\nu_o}(X, Y) = \int_{\gamma'} g^3 j(\nu_e) dl_{prop} , \quad (3.20)$$

where the infinitesimal proper length (dl_{prop}) of the emitter in the rest frame can be define as,

$$dl_{prop} = -k_\alpha u_e^\alpha d\lambda, \quad (3.21)$$

where k_α and u_e^α are the null-like and timelike four velocities of the emitter respectively. λ is the affine parameter and $g = \nu_o/\nu_e$, is the gravitational red-shift factor defined by,

$$g = \frac{k_\alpha u_o^\alpha}{k_\beta u_e^\beta}. \quad (3.22)$$

The subscript 0 and e represents the observed and emitted frequency respectively. Note that in the above equation (3.20), the integration is carried out along the null-like trajectory γ' . One can write the four velocity components of the radially freely falling accretion matter for general spherically symmetric and static spacetime as,

$$u_e^t = \frac{1}{g_{tt}(r)}, \quad u_e^r = -\sqrt{\frac{(1 - g_{tt}(r))}{g_{tt}(r)g_{rr}(r)}}, \quad u_e^\theta = u_e^\phi = 0, \quad (3.23)$$

using these four velocity components we can define the gravitational red-shift factor as,

$$g = \frac{1}{\frac{1}{g_{tt}(r)} - \frac{k_r}{k_t} \sqrt{\frac{(1 - g_{tt}(r))}{g_{tt}(r)g_{rr}(r)}}}, \quad (3.24)$$

where

$$\frac{k^r}{k^t} = \sqrt{\frac{g_{tt}(r)}{g_{rr}(r)} \left(1 - \frac{g_{tt}(r)b^2}{r^2} \right)}. \quad (3.25)$$

Now, the intensity distribution in the plane of observer's sky (X, Y) can be obtain by integrating equation (3.20) for all observed frequencies as [shaikh2019shadows],

$$I_o(X, Y) \propto - \int_{\gamma'} \frac{g^3 k_t dr}{k^r r^2}, \quad (3.26)$$

where $X^2 + Y^2 = b^2$. The above equation (3.26) represents the observed intensity of the light coming from the thin accretion matter around the compact object. Which we can use to simulate the shadow of a compact object in the observer's sky. This equation can also show how the intensity varies with the impact parameter b .

Shadow for Schwarzschild Black Hole

We can define shadow for Schwarzschild black hole by finding the effective potential of the Schwarzschild metric as we did for the general metric before.

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{\left(1 - \frac{2M}{r}\right)} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (3.27)$$

Now defining energy and angular momentum killing vector from the symmetry of time t and azimuthal coordinate ϕ .

$$\gamma = \left(1 - \frac{2M}{r}\right) \frac{dt}{d\lambda} \quad (3.28)$$

$$h = r^2 \frac{d\phi}{d\lambda} \quad (3.29)$$

Thus introducing this killing vectors in the Schwarzschild metric considering equatorial condition for constant $\theta = \frac{\pi}{2}$ and differentiating metric with respect to affine parameter λ ,

$$0 = -\frac{e^2}{\left(1 - \frac{2M}{r}\right)} + \frac{1}{\left(1 - \frac{2M}{r}\right)} \left(\frac{dr}{d\lambda}\right)^2 + \frac{l^2}{r^2} \quad (3.30)$$

from which we can write,

$$\frac{e^2}{l^2} = \frac{1}{r^2} \left(\frac{dr}{d\lambda}\right)^2 + \frac{1}{r^2} \left(1 - \frac{2M}{r}\right) \quad (3.31)$$

This is the expression of total energy of the black hole for the case of Schwarzschild metric. Last term is the expression of the effective potential,

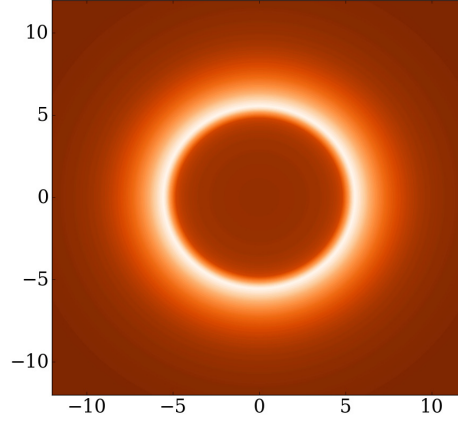


Figure 3.2: Shadow for Schwarzschild Black Hole

$$V_{eff} = \frac{1}{r^2} \left(1 - \frac{2M}{r} \right) \quad (3.32)$$

Thus from this equation, one can draw the plot of potential energy as a function of the radial distance.

Here from the graph given above, it can be interpreted that depending on the value of $b = \frac{l}{e}$, light rays will either plunge into the black hole or will get deflected and escape to infinity or will go unaffected.

From this information, we can get the shadow of the Schwarzschild black hole by plotting light rays which can be seen in 3.2, the image is taken from [shaikh2019shadows].

It is important to note that, the trajectories of the light depend just on the ratio of e and l , and thus impact parameter $b = \frac{l}{e}$.

While in the case of massive particles, the trajectories depend on the individual values of angular momentum l and energy per unit mass e of the particles.

The Schwarzschild metric represents a non-rotating the black hole. However, it is

generally believed that rotation is an important parameter to describe astrophysical black holes. Thus in the next topic, we discuss details about rotating compact objects.

3.6 Rotating Spacetime

A rotating compact object can be described by a general axisymmetric and stationary spacetime. The spacetime metric components in such spacetimes are independent of t and ϕ . This suggests that the line element is invariant to simultaneous inversion of time and azimuthal coordinates and their transformations which can be shown as,

$$t \rightarrow -t$$

$$\phi \rightarrow -\phi$$

Physical interpretation of this condition suggests a rotating spacetime geometrical structure. The invariance of the spacetime requires,

$$g_{01} = g_{02} = g_{13} = g_{23} = 0$$

as the metric components would change under the simultaneous inversion of t and ϕ .

The line element of any general spacetime metric describing a rotating compact object can be written as,

$$ds^2 = -g_{tt}dt^2 + g_{rr}dr^2 + g_{\theta\theta}d\theta^2 + g_{\phi\phi}d\phi^2 + 2g_{t\phi}dtd\phi \quad (3.33)$$

In the above metric, the term $g_{t\phi}$ corresponds rotation of the geometry.

Shadow for General symmetric metric

For the above general axisymmetric rotating spacetime, conserved quantities of energy per photon e and angular momentum per photon l can be defined as,

$$e = -g_{tt} \frac{dt}{d\lambda} - g_{t\phi} \frac{d\phi}{d\lambda} \quad (3.34)$$

$$l = g_{\phi\phi} \frac{d\phi}{d\lambda} + g_{t\phi} \frac{dt}{d\lambda} \quad (3.35)$$

From this, one can write the expressions of four velocities corresponding to t and ϕ coordinates respectively as,

$$U^t = \frac{eg_{\phi\phi} + lg_{t\phi}}{g_{t\phi}^2 - g_{tt}g_{\phi\phi}} \quad (3.36)$$

$$U^\phi = -\frac{eg_{t\phi} + lg_{tt}}{g_{t\phi}^2 - g_{tt}g_{\phi\phi}} \quad (3.37)$$

From the above expressions and the line element cooresponding to null geodesics can be written as,

$$\begin{aligned} 0 = & -g_{tt} \left(\frac{eg_{\phi\phi} + lg_{t\phi}}{g_{t\phi}^2 - g_{tt}g_{\phi\phi}} \right)^2 + g_{rr}U^{r2} + g_{\phi\phi} \left(-\frac{eg_{t\phi} + lg_{tt}}{g_{t\phi}^2 - g_{tt}g_{\phi\phi}} \right)^2 \\ & + 2g_{t\phi} \left(\frac{eg_{\phi\phi} + lg_{t\phi}}{g_{t\phi}^2 - g_{tt}g_{\phi\phi}} \right) \left(-\frac{eg_{t\phi} + lg_{tt}}{g_{t\phi}^2 - g_{tt}g_{\phi\phi}} \right) \end{aligned}$$

Here we consider the motion of null geodesics to be confined only in the equitroial plane for constant $\theta = \frac{\pi}{2}$ for simplicity. From the above equation, we can define the effective potential for a general rotating spacetime as,

$$V_{eff} = \frac{e^2g_{\phi\phi} + l^2g_{tt} + 2elg_{t\phi}}{g_{t\phi}^2 - g_{tt}g_{\phi\phi}} \quad (3.38)$$

The above equation can be used to define the trajectories of the null geodesics and the information regarding which geodesics will plunge, get deflected, and escape

to infinity and which will remain undeflected etc can be obtained.

3.7 General formalism of the shadow shapes for rotating spacetimes

In this section, we review the general formalism of the null geodesic using the Hamilton-Jacobi separation method and obtained the general expressions for finding the shape and nature of a shadow. In general relativity, the Hamilton-Jacobi equation is given by,

$$\frac{\partial S}{\partial \lambda} + \mathcal{H} = 0, \quad \mathcal{H} = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu, \quad (3.39)$$

where λ is an affine parameter, S is the Jacobi action, $\mathcal{H}$ is the Hamiltonian, and p_μ is the momentum defined by,

$$p_\mu = \frac{\partial S}{\partial x^\mu} = g_{\mu\nu} \frac{dx^\nu}{d\lambda}. \quad (3.40)$$

In the above equation (3.39), the Hamiltonian does not depend explicitly on t and ϕ coordinates. Therefore we have two constants of motion, the conserved energy $E = -p_t$ and the conserved angular momentum $L = p_\phi$ (about the axis of symmetry). Since we have the separable solution of the above differential equation (3.39), the Jacobi action can be written in terms of already known constants of the motion as,

$$S = \frac{1}{2} \mu^2 \lambda - Et + L\phi + S_r(r) + S_\theta(\theta), \quad (3.41)$$

where μ is the rest mass of the test particle. Therefore, for a photon or null geodesic, we consider $\mu = 0$. The metric tensor components of general rotating spacetime (??) are given as,

$$g^{tt} = -\frac{H - a^2 G \sin^2 \theta + 2a^2 \sqrt{\frac{G}{F}} \sin^2 \theta}{P}, \quad (3.42)$$

$$g^{rr} = \frac{FH + a^2 \sin^2 \theta}{H}, \quad (3.43)$$

$$g^{\theta\theta} = \frac{1}{H}, \quad (3.44)$$

$$g^{\phi\phi} = -\frac{G \csc^4 \theta}{2a^2 G^2 - 4a^2 G \sqrt{\frac{G}{F}} + a^2 \left(\frac{G}{F}\right) - GH \csc^2 \theta}, \quad (3.45)$$

$$g^{t\phi} = g^{\phi t} = \frac{a \left(\sqrt{\frac{G}{F}} - G \right)}{P}, \quad (3.46)$$

where,

$$P = -GH + 2a^2 G^2 \sin^2 \theta - 4a^2 G \sqrt{\frac{G}{F}} \sin^2 \theta + a^2 \left(\frac{G}{F}\right) \sin^2 \theta. \quad (3.47)$$

Substituting eq. (3.41) into eq. (3.39), and using the components of the metric tensor, we obtain the following expression:

$$\begin{aligned} - (FH + a^2 \sin^2 \theta) \left(\frac{dS_r}{dr} \right)^2 + \frac{\left(\left(\sqrt{\frac{F}{G}} H + a^2 \sin^2 \theta \right) E - aL \right)^2}{FH + a^2 \sin^2 \theta} - (L - aE)^2 \\ = \left(\frac{dS_\theta}{d\theta} \right)^2 + L^2 \cot^2 \theta - a^2 E^2 \cos^2 \theta. \end{aligned} \quad (3.48)$$

Note that, left and right hand side of above equation (3.48) are only functions of r and θ respectively, therefore one can write each side equal to a separation constant since Jacobi action principle is separable solution,

$$- (FH + a^2 \sin^2 \theta) \left(\frac{dS_r}{dr} \right)^2 + \frac{\left(\left(\sqrt{\frac{F}{G}} H + a^2 \sin^2 \theta \right) E - aL \right)^2}{FH + a^2 \sin^2 \theta} - (L - aE)^2 = \mathcal{K}, \quad (3.49)$$

$$\left(\frac{dS_\theta}{d\theta} \right)^2 + L^2 \cot^2 \theta - a^2 E^2 \cos^2 \theta = \mathcal{K}, \quad (3.50)$$

where the separation constant $\mathcal{K}$ is known as the Carter constant. Using above differential equations one can write down the following separated null geodesic equations

for radial (r) and tangential (θ) part as,

$$H^2 \left(\frac{dr}{d\lambda} \right)^2 - R(r) = 0 , \quad (3.51)$$

$$H^2 \left(\frac{d\theta}{d\lambda} \right)^2 - \Theta(\theta) = 0 , \quad (3.52)$$

where, $R(r)$ and $\Theta(\theta)$ can be considered as radial and tangential part of effective potentials respectively, which can be defines as,

$$R(r) = [X(r) - a\xi]^2 - \Delta(r) [\eta + (\xi - a)^2] , \quad (3.53)$$

$$\Theta(\theta) = \eta + (\xi - a)^2 - \left(\frac{\xi}{\sin \theta} - a \sin \theta \right)^2 \quad (3.54)$$

where, $\xi = \frac{L}{E}$ and $\eta = \frac{K}{E^2}$. The functions $X(r)$ and $\Delta(r)$ can be define as,

$$X(r) = \sqrt{\frac{F}{G}} H + a^2 \sin^2 \theta, \quad (3.55)$$

$$\Delta(r) = FH + a^2 \sin^2 \theta. \quad (3.56)$$

Note that, $R(r)$ and $\Theta(\theta)$ must be non-negative for photon motion. In a general rotating spacetime, unstable circular orbits exist when the following conditions hold,

$$R(r_{ph}) = 0, \quad \frac{dR(r_{ph})}{dr} = 0, \quad \frac{d^2 R(r_{ph})}{dr^2} \leq 0 . \quad (3.57)$$

where, $r = r_{ph}$ is the radius of the unstable photon orbit. Using the above conditions (3.57), we obtain the critical impact parameters corresponding to the maxima of the $R(r)$ as,

$$\xi = \frac{X_{ph} \Delta'_{ph} - 2 \Delta_{ph} X'_{ph}}{a \Delta'_{ph}}, \quad (3.58)$$

$$\eta = \frac{4a^2 X'_{ph} \Delta_{ph} - [(X_{ph} - a^2) \Delta'_{ph} - 2X'_{ph} \Delta_{ph}]^2}{a^2 \Delta_{ph}'^2}, \quad (3.59)$$

where prime denoted for derivative concerning r and subscript “ph” denoted for the quantities which are evaluated at $r = r_{ph}$. The above expressions are the general

forms of the critical impact parameters ξ and η for the unstable photon orbits, which would be important to find out the shape of the shadows. However, one can obtain the apparent shape of the shadow by using the celestial coordinates α and β which lie in the line of sight of the observer direction. The expressions of the celestial coordinates α and β can be written in the following way,

$$\alpha = \lim_{r_0 \rightarrow \infty} \left(-r_0^2 \sin \theta_0 \frac{d\phi}{dr} \right), \quad (3.60)$$

$$\beta = \lim_{r_0 \rightarrow \infty} \left(r_0^2 \frac{d\theta}{dr} \right), \quad (3.61)$$

where r_0 and θ_0 are the coordinates of the asymptotic observer. Note that, if we consider the general rotating metric to be an asymptotically flat in the limit of $r \rightarrow \infty$, then we can consider $G \rightarrow 1$, $F \rightarrow 1$, $H \rightarrow r^2$, $X \rightarrow r^2$ and $\Delta \rightarrow r^2$. Therefore, the above expressions of the celestial coordinates α and β becomes,

$$\alpha = -\frac{\xi}{\sin \theta_0}, \quad (3.62)$$

$$\beta = \pm \sqrt{\eta + a^2 \cos^2 \theta_0 - \xi^2 \cot^2 \theta_0}. \quad (3.63)$$

3.8 The Kerr spacetime

Using the above NJA, we can get the stationary, axisymmetric and rotating Kerr spacetime in Boyer-Lindquist coordinates as,

$$ds^2 = -\left(1 - \frac{r_s r}{\Sigma}\right) dt^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \left(r^2 + a^2 + \frac{r_s r a^2 \sin^2 \theta}{\Sigma}\right) \sin^2 \theta d\phi^2 - \frac{2r_s r a \sin^2 \theta}{\Sigma} dt d\phi, \quad (3.64)$$

where $r_s = 2M$ and

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad (3.65)$$

$$\Delta = r^2 + a^2 - r_s r. \quad (3.66)$$

The spin parameter a in length dimension is define corresponding to the total angular momentum J as $a = J/M$. Although, the dimensionless spin parameter $\tilde{a}$ can be write as $\tilde{a} = J/M^2$, where $G = c = 1$. The horizons are determine by the relation $g_{rr} \rightarrow \infty$, which implies that the solutions of $\Delta = 0$ can give us the positions of the horizons as,

$$r_{\pm} = M \pm \sqrt{M^2 - a^2} . \quad (3.67)$$

where $\pm$ indicate for the event horizon and Cauchy horizon respectively. Note that both these horizons are coincide at $r = M$ for $a = M$, which is known as extreme Kerr black hole. On the other hand, when we consider $a > M$ then the horizons are disappear and hence the Kerr black hole becomes a timelike Kerr naked singularity.

3.8.1 Frame Dragging Effect in Rotating Geometry

The rotating spacetime geometries involve $g_{t\phi}$ component in the metric tensor. Presence of this component corresponds to effect of rotation in the spacetime structure. Metric components of rotating spacetime are independent of ϕ as the structure is axisymmetric. Thus p_{ϕ} is conserved along the geodesics. We can write covariant components of angular momentum from (7.16) and (6.19) as,

$$p_t = g_{tt}p^t + g_{t\phi}p^{\phi}, \quad (3.68)$$

$$p_{\phi} = g_{\phi\phi}p^{\phi} + g_{t\phi}p^t, \quad (3.69)$$

Considering that particle has zero angular momentum i.e. $p^{\phi} = 0$, the above expressions reduces to,

$$\frac{d\phi}{dt} = \frac{p_{\phi}}{p_t} = \frac{g_{t\phi}}{g_{tt}} = \omega(r, \theta), \quad (3.70)$$

This suggests that any particle freely falling in the geodesic without any angular velocity would have a coordinate angular velocity. This effect is known as the frame-dragging effect of the inertial frame[hobson2006general].In rotating com-

pact object spacetimes such as Kerr black holes or rotating regular black holes, this frame dragging effect can affect the shadow shape of the corresponding compact object because rotation affects the trajectories of the null geodesics in the vicinity of the singularity.

In this chapter, we discussed about Schwarzschild black holes and rotating black holes and some details about their shadow structures. However, black holes are just one of the solutions to the Einstein's field equations. There are other solutions as well that might exist in reality. In the next chapter, we discuss about naked singularities.

Chapter 4

Naked singularities and Physical Properties

The theoretical study of gravitational collapse has played a central role in understanding the formation and evolution of compact objects in the universe. The foundational model for collapse of a spherically symmetric, homogeneous dust cloud was introduced by Oppenheimer, Snyder, and Dutt in the 1930s. In their analysis, known as the Oppenheimer–Snyder–Dutt (OSD) model, a massive star composed of non-interacting dust was shown to inexorably collapse under its own gravity, ultimately forming a black hole. The OSD scenario demonstrates that as the collapse proceeds, the development of trapped surfaces precedes the formation of the central space-like singularity. These trapped surfaces provide a local condition for the appearance of an event horizon: the singularity becomes hidden from the rest of spacetime, meaning that no signals can escape its vicinity once the horizon has formed.

Yet, the OSD model is a highly idealized description. It makes the simplifying assumptions that density is perfectly homogeneous and that pressure is absent throughout the collapsing star. Astrophysical objects are rarely so straightforward: their mass distributions vary spatially, and pressure gradients naturally arise as collapse proceeds. Recognizing this, researchers have pursued more realistic models by introducing inhomogeneous density profiles—these models typically assume higher

densities near the star’s core—and by allowing for non-zero pressure, which is crucial in counteracting gravitational forces and shaping the overall collapse dynamics.

When the collapsing cloud is modeled with a higher density at the center and radial variations outward, it better reflects the actual structure of massive stars. Early and contemporary studies have demonstrated that such inhomogeneity can radically change the fate of collapse. Under certain conditions, the core singularity formed during collapse may be locally or even globally visible, rather than cloaked by a horizon. These configurations are described as “naked singularities,” and their investigation tests our understanding of the limits of general relativity, most notably the cosmic censorship conjecture.

One mechanism by which naked singularities arise involves the introduction of anisotropic stresses—specifically, non-zero tangential pressure with vanishing radial pressure within the cloud. Analytical work, such as that in [joshi2011equilibrium], has explored the equilibrium configurations of such fluids, revealing that finite tangential pressure can delay or prevent the formation of trapped surfaces. This absence leads to a scenario in which the collapse process reaches a singularity visible to the outside universe, providing a theoretical counterexample to the idea that singularities must always remain hidden.

The significance of naked singularity solutions is vast: not only do they challenge the conventional view of black hole formation, but they also open exciting possibilities for observational tests in gravitational physics and astrophysics. To elucidate these ideas, this chapter presents a detailed analysis of the two principal naked singularity spacetimes—JMN-1 and Janis–Newman–Winicour (JNW)—and explores their shadow properties, which may be accessible to high-precision astronomical observations. Careful comparison with black hole shadows highlights important ways compact objects can be distinguished.

—

4.1 JMN-1 Spacetime

The JMN-1 solution describes a spacetime where naked singularity formation is permitted under physically motivated conditions. The line element for this spacetime is

$$ds^2 = -(1 - M_0) \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} dt^2 + \frac{1}{(1 - M_0)} dr^2 + r^2 d\Omega^2, \quad (4.1)$$

Here, $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$ encapsulates the two-sphere angular part, M_0 (with $0 < M_0 < 4/5$) is a dimensionless parameter regulating the central density, and R_b is the boundary radius of the matter distribution.

To accurately model the entire astrophysical object, the high-density region described by the JMN-1 metric must be matched to an external vacuum solution. At the radius $r = R_b$, the interior is joined to a Schwarzschild exterior:

$$ds^2 = - \left(1 - \frac{M_0 R_b}{r} \right) dt^2 + \frac{dr^2}{\left(1 - \frac{M_0 R_b}{r} \right)} + r^2 d\Omega^2, \quad (4.2)$$

Here, the Schwarzschild radius $R_s = M_0 R_b$ and mass $M = \frac{M_0 R_b}{2}$ arise naturally from the matching conditions.

The mathematical continuity of the two regions—the inner JMN-1 solution and the outer Schwarzschild vacuum—is enforced by standard junction conditions in general relativity. These require the extrinsic curvature to be continuous across the hypersurface at $r = R_b$, and equality of the induced metrics. Importantly, due to the vanishing radial pressure in the JMN-1 spacetime, these conditions can be satisfied without introducing additional physical inconsistencies. The model also ensures that all energy conditions—criteria on energy density and pressure profiles—hold throughout the structure.

Shadow for JMN-1 Spacetime

One of the most illuminating probes of compact object geometry comes from studying how light propagates near these objects. In the JMN-1 spacetime, it is possible

to construct the effective potential governing photon trajectories and thus predict the behavior of shadows and photon spheres. The metric takes the form:

$$ds^2 = -(1 - M_0) \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} dt^2 + \frac{dr^2}{1 - M_0} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (4.3)$$

This spacetime admits symmetries resulting in two Killing vectors—one timelike and one azimuthal—which correspond to conservation of energy and angular momentum for photons:

$$\gamma = (1 - M_0) \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} \frac{dt}{d\lambda} \quad (4.4)$$

$$h = r^2 \sin^2 \theta \frac{d\phi}{d\lambda} \quad (4.5)$$

By restricting motion to the equatorial plane, $\theta = \frac{\pi}{2}$, and differentiating the line element with respect to the affine parameter λ , we derive a first integral describing photon motion:

$$0 = -\frac{e^2}{(1 - M_0) \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}}} + \frac{1}{1 - M_0} \left(\frac{dr}{d\lambda} \right)^2 + \frac{l^2}{r^2} \quad (4.6)$$

This equation can be rewritten in terms of the impact parameter $b = l/e$, expressing a balance between centrifugal and gravitational effects:

$$\frac{1}{b^2} = \frac{e^2}{l^2} = \frac{1}{l^2} \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} \left(\frac{dr}{d\lambda} \right)^2 + \frac{1}{r^2} (1 - M_0) \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} \quad (4.7)$$

Here, the effective potential for photons is

$$V_{eff} = \frac{1}{r^2} (1 - M_0) \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} \quad (4.8)$$

This function encodes critical information: by locating the extrema of V_{eff} (i.e., $\frac{d(V_{eff})}{dr} = 0$), one finds the conditions for circular photon orbits:

$$\frac{d(V_{eff})}{dr} = \frac{d}{dr} \left[\frac{1}{r^2} (1 - M_0) \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} \right] \quad (4.9)$$

Explicit calculation gives

$$\frac{d(V_{eff})}{dr} = \frac{3M_0 - 2}{r^3} \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} \quad (4.10)$$

Therefore, for $M_0 = \frac{2}{3}$, there exists a circular photon orbit (the photon sphere) at $1/b^2 = V_{eff}$:

$$\frac{1}{b^2} = \frac{1}{r^2} \left(1 - \frac{2}{3} \right) \left(\frac{r}{R_b} \right)^{\frac{2}{3(1-\frac{2}{3})}} \quad (4.11)$$

Plugging in values yields

$$\frac{1}{b^2} = \frac{1}{3R_b^2} \quad (4.12)$$

Given $M_0 R_b = 2M$ (from matching with Schwarzschild exterior), we find $R_b = 3M$ so $b^2 = 27M^2$.

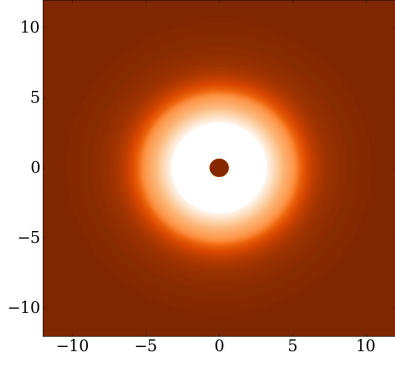
A remarkable outcome is that the photon sphere radius and the impact parameter for JMN-1 match those for the Schwarzschild black hole. That is, the photosphere appears at the same $r = 3M$. However, these photon orbits are inherently unstable: for $1/b^2 < 1/27M^2$, photons escape to infinity, while for $1/b^2 > 1/27M^2$, they are captured by the singularity.

The computed images in [4.1a](#) and

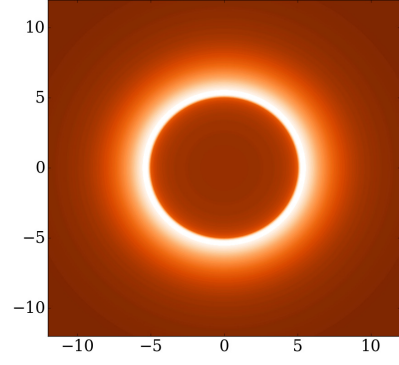
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4.2 JNW Spacetime

The Janis–Newman–Winicour (JNW) solution generalizes the Schwarzschild metric by introducing a massless scalar field. This addition drastically alters the spacetime’s global properties: the familiar event horizon is eliminated, exposing the central



(a) $R_b > 3M$: Full Moon Image for JMN-1 spacetime



(b) $R_b \leq 3M$: Shadow for JMN-1 spacetime

singularity to external observers and thus realizing a naked singularity. The JNW metric takes the form

$$ds^2 = - \left(1 - \frac{b}{r}\right)^\nu dt^2 + \left(1 - \frac{b}{r}\right)^{-\nu} dr^2 + r^2 \left(1 - \frac{b}{r}\right)^{1-\nu} d\Omega^2, \quad (4.13)$$

where

$$d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2, \quad (4.14)$$

$$\nu = \frac{2M}{b}, \quad (4.15)$$

$$b = 2\sqrt{M^2 + q^2}, \quad (4.16)$$

M is the ADM mass, q the scalar charge, and b a composite parameter. The scalar field characterizing the metric is

$$\phi = \frac{q}{b\sqrt{4\pi}} \ln \left(1 - \frac{b}{r}\right) \quad (4.17)$$

As $q \rightarrow 0$, the spacetime reduces to Schwarzschild and $\nu \rightarrow 1$. The singularity at $r = b$ is naked for $0 < \nu < 1$ and non-zero q .

—

Shadow for JNW Spacetime

The JNW metric possesses time and azimuthal independence, permitting the definition of conserved quantities for photon motion. By considering the equatorial plane ($\theta = \frac{\pi}{2}$), these are

$$\gamma = \left(1 - \frac{b}{r}\right)^n \frac{dt}{d\lambda} \quad (4.18)$$

$$h = r^2 \left(1 - \frac{b}{r}\right)^n \frac{d\phi}{d\lambda} \quad (4.19)$$

The radial dynamics are encoded in the equation

$$0 = -\frac{e^2}{\left(1 - \frac{b}{r}\right)^n} + \frac{1}{\left(1 - \frac{b}{r}\right)^n} \left(\frac{dr}{d\lambda}\right)^2 + \frac{l^2}{r^2} \left(1 - \frac{b}{r}\right)^n \quad (4.20)$$

From this, one finds the energy condition for circular photon orbits

$$\frac{1}{b^2} = \frac{e^2}{l^2} = \frac{1}{l^2} \left(\frac{dr}{d\lambda}\right)^2 + \frac{1}{r^2} \left(1 - \frac{b}{r}\right)^{3n-2} \quad (4.21)$$

yielding the effective potential

$$V_{eff} = \frac{1}{r^2} \left(1 - \frac{b}{r}\right)^{3n-2} \quad (4.22)$$

Setting the derivative to zero,

$$\frac{d(V_{eff})}{dr} = \frac{d}{dr} \left[\frac{1}{r_0^2} \left(1 - \frac{b}{r_0}\right)^{3n-2} \right] \quad (4.23)$$

Solving for r_0 yields $r_0 = \frac{3}{2}nb$. Varying n and b determines where photon spheres and shadows form in JNW geometries, and they may thus serve as indicators for the presence of a naked singularity as opposed to a black hole.

4.3 Comparison of Shadows: Black Holes versus Naked Singularities

A key outcome of the preceding analysis is the distinction in shadow properties between black holes and naked singularities. Schwarzschild black holes support a persistent photon sphere, resulting in a dark shadow observable at a fixed radius. In contrast, the existence and properties of shadows in naked singularity spacetimes—for example JMN-1 or JNW—depend sensitively on spacetime parameters. The JMN-1 metric yields a photon sphere (hence a shadow) if $M > \frac{2}{3}$. For $M < \frac{2}{3}$, no photon sphere or shadow exists, allowing outgoing photons to escape without encountering a critical orbit.

Shadows therefore serve as a powerful diagnostic: their presence, location, and size can provide clues to the underlying spacetime geometry. Nevertheless, shadow observations alone may not suffice for absolute identification. Complimentary signatures such as the behavior of timelike geodesics, orbital precession rates, or accretion flow properties could provide additional evidence to distinguish black holes from naked singularities in astrophysical settings.

On the Possibility of Compact Objects Without Singularities

All spacetimes discussed thus far—both black holes and naked singularities—feature a central curvature singularity, signaled by divergent metric components at a critical radius. This behavior stems from the form of the metric and the underlying energy conditions. The prospect remains, however, for compact objects where singularity formation does not occur, either due to modification of the energy conditions or via quantum gravitational effects. The nature and characterization of such singularity-free compact objects will be the focus of the next chapter.

Chapter 5

Timelike orbits in compact objects

5.1 Timelike orbits and precession

In the previous chapters, we have discussed about null geodesics in compact object spacetimes of black holes and naked singularities. It is usually believed that there are compact objects at centre of different galaxies. In case of our Milky Way too, the scenario is very similar. Many specific stars are rotating around the centre of our galaxy. The motion and trajectories of these stars are used as a potential signature to distinguish or probe the nature of the compact object at the galactic centre. Thus it is important to study the timelike geodesics of compact objects as well along with the null geodesics. In this chapter, we discuss timelike geodesics for general spacetimes and then for Schwarzschild black hole, JMN-1 naked singularity and JNW naked singularity.

5.2 Timelike geodesic in spherically symmetric and static spacetime

The above mentioned Schwarzschild spacetime (3.4) is spherically and temporally symmetric. Thus the angular momentum and energy of a particle per unit rest mass are always conserved. These two quantities are important to study and investigate

the nature of timelike orbits. The conserved angular momentum per unit rest mass (h) and conserved energy per unit rest mass (γ) for a freely falling particle in the spherically symmetric and static spacetime can be written as,

$$h = g_{\phi\phi}(r) \frac{d\phi}{d\tau} , \quad \gamma = g_{tt}(r) \frac{dt}{d\tau} , \quad (5.1)$$

here τ denotes proper time of the particle. We consider the normalization condition for the derivation of orbit equation of timelike particles $u^\mu u_\mu = -1$, where u^μ is four velocity of any massive particle. Now using the conservation laws given in (5.1) and normalization of four velocity relation, general expression of energy for a freely falling particle in a general spherically symmetric spacetime can be written as,

$$E = \frac{g_{rr}(r)g_{tt}(r)}{2} \left(\frac{dr}{d\tau} \right)^2 + V_{eff}(r) , \quad (5.2)$$

where $E = \frac{\gamma^2 - 1}{2}$. Here we consider an equatorial plane ($\theta = \frac{\pi}{2}$) for the particle's trajectories for simplicity. The expression of radial effective potential can be written as,

$$V_{eff}(r) = \frac{1}{2} \left[g_{tt}(r) \left(1 + \frac{h^2}{g_{\phi\phi}(r)} \right) - 1 \right] , \quad (5.3)$$

The above equation of effective potential plays an important role on the motion of the particles in a given general form of spacetime (3.4). The above mentioned two quantities, namely total energy and effective potential follow certain relation which are,

$$V_{eff}(r) = E, \quad V'_{eff}(r) = 0, \quad (5.4)$$

The above expressions can be useful to get different types of bound orbital shapes for examples, circular orbits and elliptical orbits. From the above mentioned conditions, the expressions conserved angular momentum (h) and energy (γ) can be derived for

circular orbits,

$$h^2 = \frac{g_{\phi\phi}(r)}{\left(\frac{2g_{tt}(r)}{\sqrt{g_{\phi\phi}(r)g'_{tt}(r)}} - 1\right)}, \quad \gamma^2 = \frac{2g_{tt}^2(r)}{\sqrt{g_{\phi\phi}(r)g'_{tt}(r)}} \frac{1}{\left(\frac{2g_{tt}(r)}{\sqrt{g_{\phi\phi}(r)g'_{tt}(r)}} - 1\right)}, \quad (5.5)$$

where prime denotes derivative with respect to r . It is important to note that the above mentioned conditions $V_{eff}(r) = E$, $V'_{eff}(r) = 0$, are valid for both stable and unstable circular orbits. The stability of the circular orbits defined by the second derivatives of effective potential with respect to r as,

$$V''_{eff}(r) > 0, \quad V''_{eff}(r) < 0, \quad (5.6)$$

for stable and unstable circular orbits respectively. One can get bound elliptical orbits by employing the condition $V_{eff}(r) = E$, in which the minimum (r_{min}) and maximum (r_{max}) radial approach of the freely falling test particle from the center needs to be established. This can be used to define necessary conditions for the existence of the stable bound orbits for freely falling test particle,

$$V_{eff}(r_{min}) = V_{eff}(r_{max}) = E, \quad (5.7)$$

$$E - V_{eff}(r) > 0, \quad \forall r \in (r_{min}, r_{max}). \quad (5.8)$$

The shape of the timelike bound orbits can describe using eq. (5.2) for particular values of conserved energy γ and angular momentum h ,

$$\frac{dr}{d\phi} = \frac{g_{\phi\phi}(r)}{h} \frac{\sqrt{2(E - V_{eff})}}{\sqrt{g_{rr}(r)g_{tt}(r)}}. \quad (5.9)$$

For the null geodesics, an expression similar to above was derived. Differentiation of the above eq. (5.9) with respect to ϕ can be used for the derivation of the orbit

equation of a massive particle as follows,

$$\frac{d^2 u}{d\phi^2} + \frac{\gamma^2 g_{\phi\phi}^2(u) u^4}{2g_{tt}^2(u)g_{rr}(u)h^2} \left(\frac{dg_{tt}(u)}{du} \right) - X(u) \left(\frac{dg_{rr}(u)}{du} \right) + Y(u) \left(\frac{dg_{\phi\phi}(u)}{du} \right) - Z(u) = 0 , \quad (5.10)$$

here $u = \frac{1}{r}$ and,

$$X(u) = \left[\frac{\gamma^2 g_{\phi\phi}^2(u) u^4}{2g_{tt}(u)g_{rr}^2(u)h^2} - \frac{g_{\phi\phi}(u) u^4}{2g_{rr}^2(u)} - \frac{g_{\phi\phi}^2(u) u^4}{2g_{rr}^2(u)h^2} \right] , \quad (5.11)$$

$$Y(u) = \left[\frac{\gamma^2 g_{\phi\phi}(u) u^4}{g_{tt}(u)g_{rr}(u)h^2} - \frac{u^4}{2g_{rr}(u)} - \frac{g_{\phi\phi}(u) u^4}{g_{rr}(u)h^2} \right] , \quad (5.12)$$

and

$$Z(u) = \left[\frac{2\gamma^2 g_{\phi\phi}^2(u) u^3}{g_{tt}(u)g_{rr}(u)h^2} + \frac{2g_{\phi\phi}(u) u^3}{g_{rr}(u)} + \frac{2g_{\phi\phi}^2(u) u^3}{g_{rr}(u)h^2} \right] . \quad (5.13)$$

It is understandable that the above equation (5.10) is a second-order non-linear differential equation. Thus it is very hard to derive an analytical solution for it. Thus it is solved numerically to investigate the nature and shape of the timelike bound orbits for massive particles. From the discussion regarding timelike orbits, it can be understood that the nature and shape of the timelike bound orbits depend on the causal structure of spacetime geometry.

5.3 Schwarzschild black hole orbit equation

The Schwarzschild spacetime is defined as given in equation (3.1). The geometry has a singularity at $r = 0$ covered by an event horizon at $r = 2M$. Since the metric components of the Schwarzschild spacetime are independent of t and ϕ , one can write the conserved energy and angular momentum per unit rest mass of the freely falling test particle in this spacetime as,

$$\gamma_{SCH} = \left(1 - \frac{2M}{r} \right) \left(\frac{dt}{d\tau} \right) , \quad (5.14)$$

$$h_{SCH} = r^2 \left(\frac{d\phi}{d\tau} \right) , \quad (5.15)$$

From the previously derived general equation (5.3), we can define the effective potential energy of massive particles in Schwarzschild spacetime as,

$$(V_{eff})_{SCH} = \frac{1}{2} \left[\left(1 - \frac{M_0 R_b}{r} \right) \left(1 + \frac{h_{SCH}^2}{r^2} \right) - 1 \right] . \quad (5.16)$$

Here $2M = M_0 R_b$. M_0 and R_b are parameters of the metric components. The expressions of the conserved energy and angular momentum for circular geodesics are defined as from the conditions $V_{eff}(r) = E$ and $V'_{eff}(r) = 0$,

$$\gamma_{SCH}^2 = \frac{2(r - M_0 R_b)^2}{r(2r - 3M_0 R_b)} , \quad (5.17)$$

$$h_{SCH}^2 = \frac{M_0 R_b r^2}{(2r - 3M_0 R_b)} \quad (5.18)$$

From the above expressions, it can be seen that no circular orbit is possible within the range $0 \leq r \leq \frac{3M_0 R_b}{2}$, or in terms of total mass of the black hole, we can write $0 \leq r \leq 3M$, where $M = \frac{3M_0 R_b}{2}$. Note that above ranges are valid for both the stable and unstable circular orbits. Therefore, for stable circular orbits we have to use the $(V''_{eff})_{SCH} > 0$ condition as,

$$(V''_{eff})_{SCH} = \frac{2M}{r} \left(\frac{6M - r}{3M - r} \right) . \quad (5.19)$$

From the above expression, one can see that no stable circular orbits are possible below $r = 6M$ in the Schwarzschild spacetime. Thus, $r = 6M$ is known as the innermost stable circular orbit (ISCO) in the Schwarzschild spacetime. Note that for bound orbits, the conditions given in equations (5.7) and (5.8) should be fulfilled.

Now, an orbit equation of the test particle can be written using the general expression of the orbit equation given in (5.10) for the Schwarzschild spacetime as,

$$\frac{d^2u}{d\phi^2} + u = \frac{3M_0R_b}{2}u^2 + \frac{M_0R_b}{2h^2} . \quad (5.20)$$

which is the second order non-linear differential equation.

5.3.1 JMN-1 orbit equation

The JMN-1 naked singularity is non-vacuum solution of Einstein equations. As we discussed before JMN-1 is formed as an end state of gravitational collapse with inhomogenous matter and presence of pressure in the stress energy tensor. It has strong curvature singularity at the center without an event horizon. As it is spherically symmetric and static solution, we can write the conserved energy and angular momentum per unit rest mass as,

$$\gamma_{JMN-1} = (1 - M_0) \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} \left(\frac{dt}{d\tau} \right) , \quad (5.21)$$

$$h_{JMN-1} = r^2 \left(\frac{d\phi}{d\tau} \right) , \quad (5.22)$$

now, employing the general formalism of the orbit equation, we can write the effective potential for the JMN-1 spacetime as,

$$(V_{eff})_{JMN-1} = \frac{1}{2} \left[(1 - M_0) \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}} \left(1 + \frac{h_{JMN-1}^2}{r^2} \right) - 1 \right] , \quad (5.23)$$

where h_{JMN-1} is the conserved angular momentum per unit rest mass of the test particles in JMN-1. The conserved quantities for circular orbits are defined as,

$$\gamma_{JMN-1}^2 = \frac{2(1 - M_0)^2 \left(\frac{r}{R_b} \right)^{\frac{M_0}{1-M_0}}}{(2 - 3M_0)} , \quad (5.24)$$

$$h_{JMN-1}^2 = \frac{r^2 M_0}{2 - 3M_0} , \quad (5.25)$$

the stable circular orbit can obtained by using condition,

$$(V''_{eff})_{JMN-1} = \frac{M_0}{R_b^2} \left(\frac{r}{R_b} \right)^{\frac{3M_0-2}{1-M_0}} > 0 \quad (5.26)$$

from the above equations it is clearly seen that for $M_0 < 2/3$, JMN-1 spacetime can have stable circular orbits of any radius. Note that in JMN-1 spacetime, the unstable circular orbits are not exist within the region $0 < M_0 < 2/3$. Since, h_{JMN-1} and γ_{JMN-1} become imaginary for $M_0 > 2/3$, the stable circular orbits are not possible at any value of r in JMN-1 spacetime. Now, an orbit equation of a freely falling test particle in JMN-1 spacetime can write as,

$$\frac{d^2u}{d\phi^2} + (1 - M_0)u - \frac{\gamma^2}{2h^2} \frac{M_0}{(1 - M_0)} \left(\frac{1}{u} \right) \left(\frac{1}{uR_b} \right)^{\frac{-M_0}{(1-M_0)}} = 0 , \quad (5.27)$$

5.3.2 JNW orbit equation

The JNW spacetime is minimally coupled mass-less scalar field solution of the Einstein equations. Unlike, JMN-1 naked singularity spacetime, JNW naked singularity spacetime is asymptotically flat. Therefore, this spacetime need not to match with the external Schwarzschild spacetime. One can see from the equation (??), the metric tensor components of this spacetime is independent of t and ϕ . Thus, we can define the conserved energy and angular momentum of a test particle per unit rest mass as,

$$h_{JNW} = r^2 \left(1 - \frac{b}{r} \right)^{1-n} \frac{d\phi}{d\tau} , \quad (5.28)$$

$$\gamma_{JNW} = \left(1 - \frac{b}{r} \right)^n \frac{dt}{d\tau} . \quad (5.29)$$

Similarly, using the general formalism, we can write the effective potential of a freely falling test particle for JNW spacetime as,

$$(V_{eff})_{JNW} = \frac{1}{2} \left[\left(1 - \frac{b}{r} \right)^n \left(1 + \frac{h_{JNW}^2}{r^2} \left(1 - \frac{b}{r} \right)^{n-1} \right) - 1 \right] , \quad (5.30)$$

the conserved quantities for circular geodesics can be defined as,

$$\gamma_{JNW}^2 = \left(1 - \frac{b}{r}\right)^n \left[\frac{2r - b(n+1)}{2r - b(2n+1)} \right] , \quad (5.31)$$

$$h_{JNW}^2 = r^2 \left[\frac{bn \left(1 - \frac{b}{r}\right)^{1-n}}{2r - b(2n+1)} \right] , \quad (5.32)$$

where the stable circular orbits would be excess by applying the condition $(V_{eff}'') > 0$. Although, In JNW spacetime, (V_{eff}'') is not always positive [parth1]. We can find $(V_{eff}'') > 0$ at any point in JNW spacetime for $n < 0.447$. But, for $0.447 < n < 0.5$, we have one certain radial interval (r_-, r_+) , in which no stable circular orbits are possible, where r_- and r_+ can be written as,

$$r_- = \frac{1}{4} \left(b(2 + 6n) - 4.472 b\sqrt{n^2 - 0.2} \right) , \quad (5.33)$$

$$r_+ = \frac{1}{4} \left(b(2 + 6n) + 4.472 b\sqrt{n^2 - 0.2} \right) . \quad (5.34)$$

Note that, the stable circular orbits are not possible in the certain radial interval $r_- < r < r_+$. However, we can have stable circular orbits outside this radial interval as there we have $(V_{eff}'')_{JNW} > 0$. On the other hand, for $0.5 < n < 1$ we can get innermost stable circular orbit (ISCO) at $r_{(ISCO)}$. Now, employing the general method to derive an orbit equation for the JNW spacetime, we can get the second order non-linear differential equation as,

$$\frac{d^2u}{d\phi^2} + u + \frac{b\gamma^2}{2h^2}(2 - 2n)(1 - bu)^{1-2n} - \frac{b(2 - n)}{2h^2}(1 - bu)^{1-n} - \frac{3bu^2}{2} = 0 . \quad (5.35)$$

Chapter 6

Wormholes and Physical Properties

There exist numerous possible solutions to Einstein's field equations, each describing distinct classes of compact astrophysical objects. So far, we have examined black holes and naked singularities, both of which exhibit spacetime singularities within their geometrical structures. However, alternative solutions with entirely different characteristics may also exist. One particularly intriguing possibility is that of the wormhole. Wormholes constitute exact solutions to Einstein's field equations and were first introduced by Einstein and Rosen. They represent tunnel-like structures in spacetime that connect two separate points in the manifold. Owing to this property, they are often referred to as Einstein–Rosen bridges. In this chapter, we undertake a detailed examination of the spacetime geometry of wormholes, the physical interpretation of their metric components, the role of energy conditions in their stability, and the phenomenon associated with their optical appearance, namely, the shadow of wormholes.

6.1 Property of wormhole

As discussed earlier, unlike black holes or naked singularities, wormholes do not contain singularities in their spacetime geometry. From the form of their line element,

it can be inferred that spacetime converges from the asymptotic region to a finite radius, attains a constant minimal surface, and subsequently diverges toward another asymptotically flat region. This continuous convergence and divergence pattern constitutes a fundamental characteristic of the wormhole spacetime, distinguishing it from other compact configurations. The spatial surface at which the spacetime curvature reaches a constant finite radius is termed the throat of the wormhole.

To comprehend such a structure from a geometrical standpoint, it becomes necessary to analyze the expansion scalar associated with the spacetime, which encapsulates information regarding the local convergence or divergence of null geodesics. In the following section, we derive and study the expansion scalar for a wormhole metric.

6.1.1 Expansion scalar

The expansion scalar, denoted by θ_l , quantifies the rate of divergence or convergence of a bundle of null geodesics and, hence, provides direct insight into the spacetime geometry. It can be formally defined in terms of the null geodesic congruence[whiting2004relativist]. For compact objects possessing central singularities, the expansion scalar diverges to infinity, indicating that geodesics converge to a single point in spacetime. Conversely, in the case of a wormhole, the scalar remains finite, signifying the absence of a singularity. Its value depends explicitly on the form of the metric components. Mathematically, it is expressed as

$$\theta_l = \nabla_\mu l^\mu + \frac{l^\mu n^\nu + n^\mu l^\nu}{-l^\alpha n_\alpha} \nabla_\mu l_\nu. \quad (6.1)$$

Expansion scalar for wormhole metric

Let us consider a simple curved spacetime[whiting2004relativist] characterized by the line element

$$ds^2 = -dt^2 + dl^2 + r(l)^2 d\theta^2 + r(l)^2 \sin^2 \theta d\phi^2. \quad (6.2)$$

In this configuration, $r(l) = 0$ corresponds to the minimum radius at $l = 0$, while $r(l) = l$ in the asymptotic region as $l \rightarrow \infty$. From this metric, one can calculate the expansion scalar θ_l to probe the geometrical nature of the object. It is given by

$$\theta_l = \nabla_\mu l^\mu + \frac{l^\mu n^\nu + n^\mu l^\nu}{-l^\alpha n_\alpha} \nabla_\mu l_\nu. \quad (6.3)$$

Consider first the case $l = 0$ and $r(l) = r_0$, for which the metric simplifies to

$$ds^2 = -dt^2 + r_0^2 d\theta^2 + r_0^2 \sin^2 \theta d\phi^2. \quad (6.4)$$

For this metric, we define

$$l^\mu = 1, 0, 0, 0, \quad (6.5)$$

$$n^\mu = 1, 0, 0, 0. \quad (6.6)$$

Hence,

$$l^\mu n_\mu = -1. \quad (6.7)$$

and

$$\nabla_\mu l^\mu = \nabla_0 l^0 + \nabla_1 l^1 + \nabla_2 l^2 + \nabla_3 l^3 = \Gamma_{00}^0 l^0 + \Gamma_{10}^1 l^0 + \Gamma_{20}^2 l^0 + \Gamma_{30}^3 l^0. \quad (6.8)$$

With

$$\nabla_\mu l_\nu = g_{\alpha\nu} \nabla_\mu l^\alpha, \quad (6.9)$$

the corresponding term in the expansion scalar becomes

$$(l^\mu n^\nu + n^\mu l^\nu)g_{\alpha\nu}\nabla_\mu l^\alpha = 2\Gamma_{00}^0 l^0. \quad (6.10)$$

Therefore, for $l = 0$, the expansion scalar reduces to

$$\theta_l = 3\Gamma_{00}^0 l^0 + \Gamma_{10}^1 l^0 + \Gamma_{20}^2 l^0 + \Gamma_{30}^3 l^0. \quad (6.11)$$

Similarly, in the asymptotic region where $l \rightarrow \infty$ and $r(l) = l$, the line element becomes

$$ds^2 = -dt^2 + dl^2 + l^2 d\theta^2 + l^2 \sin^2 \theta d\phi^2. \quad (6.12)$$

The respective ingoing and outgoing null geodesics are expressed as

$$l^\mu = 1, 1, 0, 0, \quad (6.13)$$

$$n^\mu = 1, -1, 0, 0, \quad (6.14)$$

which yield

$$l^\mu n_\mu = g_{\mu\nu} l^\mu n^\nu = -2. \quad (6.15)$$

and

$$\nabla_\mu l^\mu = \nabla_0 l^0 + \nabla_1 l^1 + \nabla_2 l^2 + \nabla_3 l^3 = \Gamma_{00}^0 l^0 + \Gamma_{01}^0 l^1 + \Gamma_{11}^1 l^1 + \Gamma_{10}^1 l^0 + \Gamma_{20}^2 l^0 + \Gamma_{21}^2 l^1 + \Gamma_{30}^3 l^0 + \Gamma_{31}^3 l^1. \quad (6.16)$$

For this case,

$$\nabla_\mu l_\nu = g_{\alpha\nu} \nabla_\mu l^\alpha, \quad (6.17)$$

and

$$(l^\mu n^\nu + n^\mu l^\nu)g_{\alpha\nu}\nabla_\mu l^\alpha = 2(\Gamma_{00}^0 l^0 + \Gamma_{01}^0 l^1 + \Gamma_{00}^1 l^0 + \Gamma_{01}^1 l^1 + \Gamma_{10}^0 l^0 + \Gamma_{11}^0 l^1 + \Gamma_{10}^1 l^0 + \Gamma_{11}^1 l^1). \quad (6.18)$$

From these relations, the expansion scalar for $l \rightarrow \infty$ becomes

$$\theta_l = 2(\Gamma_{00}^0 + \Gamma_{01}^0 + \Gamma_{10}^1 + \Gamma_{11}^1) + \Gamma_{00}^1 + \Gamma_{01}^1 + \Gamma_{10}^0 + \Gamma_{11}^0 + \Gamma_{20}^2 + \Gamma_{21}^2 + \Gamma_{30}^3 + \Gamma_{31}^3. \quad (6.19)$$

Upon explicit evaluation of the Christoffel symbols, we find the following results:

(1) For $l = 0$ and $r(l) = r_0$,

$$\theta_l = 0. \quad (6.20)$$

This indicates that at $l = 0$, the spacetime neither converges nor diverges but maintains a finite radius r_0 . This surface corresponds to the throat of the wormhole.

(2) In the limit $l \rightarrow \infty$, as $r(l) = l$,

$$\theta_l = \frac{2}{l}. \quad (6.21)$$

Hence, as one moves inward from the asymptotic region, the spacetime converges until the throat is reached, and beyond it the spacetime diverges. Thus, through the expansion scalar derived from the metric, the nature and structure of the wormhole spacetime can be fully characterized, confirming the absence of any curvature singularity. The subsequent section explores various wormhole spacetimes and their corresponding physical properties.

6.2 Wormhole Spacetimes and Corresponding Physical Properties

The precise mechanism underlying the formation of wormholes remains an outstanding open question in theoretical physics. Nevertheless, a number of spacetime metrics have been constructed to model traversable wormhole geometries. These metrics are typically derived through one of two approaches[**morris1988wormholes**]:

(1) In the first approach, the spacetime metric is deliberately fashioned so that its components satisfy the defining criteria for a traversable wormhole, such as those articulated through the expansion scalar analysis. Subsequent investigations then focus on the required matter distribution, deduced via Einstein’s field equations, to support such geometries.

(2) Alternatively, researchers may begin by positing a particular matter distribution and employ Einstein’s field equations to infer the corresponding metric structure.

This dual framework illustrates that, provided either the geometrical prescription or the energy–momentum distribution is specified, the remaining physical characteristics of the wormhole spacetime can be systematically obtained.

Ellis Wormhole

The Ellis wormhole spacetime, representing a particular instance of the Ellis drain-hole, exemplifies a purely traversable geometry devoid of horizons and singularities. Its explicit line element is given by[**ohgami2015wormhole**]:

$$ds^2 = -dt^2 + dr^2 + (r^2 + a^2)d\theta^2 + (r^2 + a^2)\sin^2\theta d\phi^2, \quad (6.22)$$

Here, the parameter a denotes the throat radius. Analysis of the metric components and the expansion scalar unequivocally demonstrates the absence of a curvature singularity. The same spacetime metric, 6.2, can equivalently be rewritten in

terms of a new radial coordinate $r^* = (r^2 + a^2)^{1/2}$ as follows:

$$ds^2 = \frac{dr^{*2}}{1 - \frac{a^2}{r^{*2}}} + r^{*2}d\phi^2, \quad (6.23)$$

To facilitate geometric visualization, consider the line element for three-dimensional Euclidean space in cylindrical coordinates:

$$ds^2 = dr^{*2} + r^{*2}d\phi^2 + dz^2. \quad (6.24)$$

Comparing equations (6.23) and (6.24), we obtain:

$$\frac{dr^{*2}}{1 - \frac{a^2}{r^{*2}}} + r^{*2}d\phi^2 = dr^{*2} + r^{*2}d\phi^2 + dz^2, \quad (6.25)$$

from which the relation

$$dz = \frac{1}{1 - \frac{a^2}{r^{*2}}} \quad (6.26)$$

follows. Integrating yields

$$z = \pm a \cosh^{-1} \left(\frac{r^*}{a} \right), \quad (6.27)$$

which provides a geometric embedding of the wormhole surface, as depicted in Figure 6.1.

Morris–Thorne Wormhole

A particularly influential class of wormhole solutions was introduced by Morris and Thorne. The corresponding spacetime metric takes the form[[kim2013flare](#)]:

$$ds^2 = -e^{2\phi(r)}dt^2 + \frac{dr^2}{1 - \frac{B(r)}{r}} + r^2d\theta^2 + r^2\sin^2\theta d\phi^2, \quad (6.28)$$

Here, $\phi(r)$ denotes the redshift function, which encodes the gravitational time dilation effects encountered by radially propagating photons. Commonly, $\phi(r)$ is

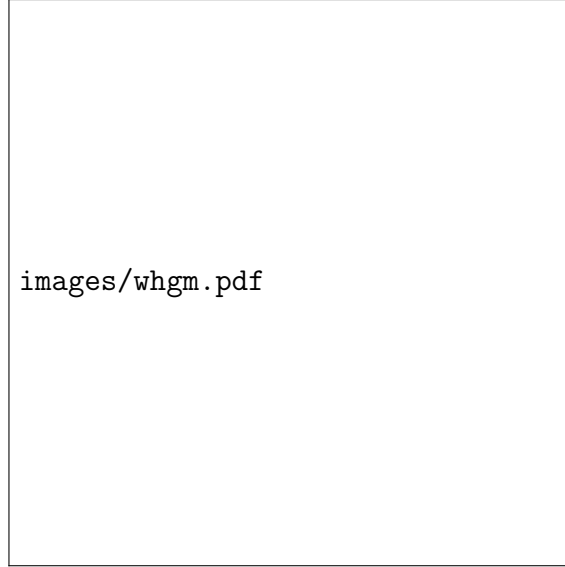


Figure 6.1: Wormhole Spacetime Geometry

parametrized as $-\frac{r_0}{r}$, where r_0 identifies the throat radius. In the Newtonian limit, r_0 can be related to the mass of the compact object:

$$e^{2\phi(r)} = e^{-\frac{r_0}{r}}, \quad (6.29)$$

which, retaining only the leading terms, reduces to

$$e^{2\phi(r)} = 1 - \frac{r_0}{r} = 1 - \frac{2M}{r}. \quad (6.30)$$

The function $B(r)$ is termed the shape function, as its specific form determines the spatial geometry of the wormhole. Detailed consideration of $B(r)$ and the conditions under which it yields a physically viable wormhole will be provided in subsequent sections.

In Morris–Thorne wormholes, the physical characteristics are governed primarily by the behavior of the redshift and shape functions—collectively referred to as generating functions[**rahaman2019generating**]. These functions impose crucial constraints that ensure the resulting metric describes a wormhole, rather than another class of compact object.

One salient property distinguishing wormholes from black holes or naked singularities is the absence of an event horizon and central singularity. This is intimately

connected to the g_{tt} component, which, via the redshift function, must remain finite and nonzero to preclude the formation of either a horizon or singular geometry. As a result, wormhole spacetimes are regular everywhere and tend toward flatness in the asymptotic limit.

Another key feature is the topological structure, whereby spacetime converges in one region, achieves a constant minimal radius at the throat, and subsequently diverges toward another region—either a distinct universe or a separate neighborhood of the same universe. This behavior is encoded in the shape function and radial coordinate. Thus, a deeper understanding of wormhole geometry requires detailed analysis of the interplay between the redshift and shape functions.

For the Morris–Thorne metric [6.28](#), specializing to $t = \text{constant}$ and $\theta = \frac{\pi}{2}$, yields

$$ds_2^2 = \frac{dr^2}{1 - \frac{B(r)}{r}} + r^2 d\phi^2. \quad (6.31)$$

By equating this reduced line element with its Euclidean counterpart (from equation [\(3.68\)](#)), we obtain

$$ds_2^2 = \frac{dr^2}{1 - \frac{B(r)}{r}} + r^2 d\phi^2 = dr^2 + r^2 d\phi^2 + dz^2. \quad (6.32)$$

Focusing solely on the radial coordinates leads to

$$1 + \left(\frac{dz}{dr}\right)^2 = \frac{1}{1 - \frac{B(r)}{r}}, \quad (6.33)$$

which can be re-expressed as

$$\left(\frac{dz}{dr}\right) = \pm \frac{1}{\left(\frac{r}{B(r)} - 1\right)^{1/2}}. \quad (6.34)$$

This relation underscores the influence of the shape function $B(r)$ on the geometrical character of the spacetime. In the vicinity of the throat, where $B(r) = r$, the derivative (dz/dr) diverges, reflecting the breakdown of the coordinate r . To

remedy this, one replaces r by the proper radial distance[**morris1988wormholes**]:

$$dl = \pm \frac{dr}{\left(\frac{r}{B(r)} - 1\right)^{1/2}} \quad (6.35)$$

Expressed in terms of l , the geometry remains regular throughout, including at the throat. Consequently, the proper choice of radial coordinate is essential for physical description of the wormhole structure.

The throat, from a physical standpoint, is defined as the minimal-area surface linking two separate regions. To represent this feature, the shape function must satisfy stringent flare-out conditions.

Flare-Out Condition

The flare-out condition[**kim2013flare**] stipulates that at the throat $r = r_0$, the shape function must satisfy $B(r_0) = r_0$, while its derivative fulfills $B'(r_0) < 1$.

The question of why singularities do not arise in wormhole spacetimes—particularly at the throat—naturally leads to consideration of stability mechanisms. The theoretical explanation for the persistence of a regular throat resides in the violation of standard energy conditions, notably the strong energy condition.

6.3 Violation of Strong Energy Conditions

Energy conditions comprise generalized constraints on the stress–energy tensor, intended to encapsulate the physical requirement that the energy density should not be negative. These conditions, while motivated by considerations of physical reasonableness, are not rigorously derivable from first principles. In the context of wormholes, it is well-established that the strong energy condition (SEC) must be violated for the maintenance of a stable wormhole configuration.

Insight into this requirement is provided by the focusing theorem[**whiting2004relativist**]. The theorem predicts that a congruence of timelike geodesics, subject to the SEC, will generically focus to a caustic point—a locus of infinite density. In contrast, the

geodesics associated with a wormhole exhibit convergence only up to a finite radius (the throat), after which they begin to diverge, indicating the absence of a focal singularity. This behavior is summarized by the Raychaudhuri equation:

$$\frac{d\theta}{d\tau} = -\frac{1}{3}\theta^2 - \sigma^{\alpha\beta}\sigma_{\alpha\beta} + \omega^{\alpha\beta}\omega_{\alpha\beta} - R_{\alpha\beta}u^\alpha u^\beta \quad (6.36)$$

For wormhole spacetimes, in order for initially converging geodesics to later diverge, it is necessary that $R_{\alpha\beta}u^\alpha u^\beta < 0$. From this analysis, it follows that the stable structure of a wormhole inherently requires violation of the strong energy condition.

Having addressed the geometric construction and energy condition requirements of wormhole spacetimes, we next turn to a potentially observable astrophysical signature—the shadow cast by a Morris–Thorne wormhole.

6.4 Shadow of Wormhole

For the Morris–Thorne wormhole metric [6.28](#), the conserved quantities for null geodesics—namely, the energy and angular momentum per photon—are given by

$$\gamma = e^{2\phi(r)} \frac{dt}{d\lambda}, h = r^2 \frac{d\phi}{d\lambda}. \quad (6.37)$$

Leveraging these conservation laws, one obtains

$$\frac{e^2}{l^2} = \frac{1}{r^2} \left(\frac{dr}{d\lambda} \right)^2 + \frac{1}{r^2} (e^{2\phi(r)}), \quad (6.38)$$

which leads directly to an expression for the effective potential for null geodesics:

$$V_{eff} = \frac{1}{r^2} (e^{2\phi(r)}). \quad (6.39)$$

The effective potential furnishes a powerful tool for analyzing the behavior of photons in the wormhole spacetime, including their deflection angles, stable and unstable orbits, and the resultant shadow features[[muller2008exact](#)]. Though exten-

sive theoretical studies have examined such observables in various wormhole configurations, the present chapter has focused on elucidating the fundamental geometric structure. In the chapter to follow, we will explore a specific class of wormhole metrics for which the effective potential and orbital dynamics have not been previously addressed in the literature.

Chapter 7

Humanly Traversable Wormhole

As discussed in the previous chapter, the structure of the wormhole is very unique as it do not possess the singularity. Apart from this, wormholes can act as geometrical structures which allows the possibility of time travel. These facts make them very interesting. There are many different articles which consider wormholes as a potential candidate of compact objects present in the universe, especially at the galactic centre. However, there has not yet been any observational signatures of existence of these compact objects. Also, the mechanism of formation of this compact objects is not understood. Violation of energy condition and requirement of exotic matter in the throat region are also important aspects that cannot be ignored while thinking about wormholes. These all reasons contribute to the perspective of existence of wormholes being questionable.

However in [**maldacena2021humanly**], the authors discuss the fact that existence of wormhole is forbidden only within the classical general theory of relativity. They are allowed in the quantum theory. Thus they propose a wormhole model with desirable properties by introducing a dark sector, which interacts with the standard model via gravity. Authors put emphasis on this type of wormholes being consistent with known laws of physics. For this study, they use the Randall-Sundrum model. Thus in this chapter we discuss the effective potential and null geodesics as well as timelike orbits around this model of wormhole.

7.1 Spacetime Metric of Humanly Traversable Wormhole

This wormhole model suggests two magnetic black hole spacetimes with opposite charges, d distance apart from each other. However, both spacetimes are connected by a throat region of l with each other rather than possessing singularities in their structures. Thus the spacetime structure of this wormhole consists of two different spacetime metric: Outer region is represented by line element of a magnetic black hole and Inner region is represented by another line element which corresponds to throat region of the wormhole.

The line element of exterior outer region of spacetime is given by,

$$ds^2 = -\left(1 - \frac{r_e}{r}\right)^2 dt^2 + \frac{dr^2}{\left(1 - \frac{r_e}{r}\right)^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2,$$

where r_e is the radius of the spacetime continuum in the region near the horizon.

While the line element for the spacetime metric of the throat region connecting two black holes is given by,

$$ds^2 = -r_e^2 (\rho^2 + 1) dt^2 + \frac{r_e^2 dr^2}{(\rho^2 + 1)} + r_e^2 d\theta^2 + r_e^2 \sin^2 \theta d\phi^2, -\rho_c < \rho < \rho_c \quad (7.1)$$

here,

$$\rho = l \frac{(r - r_e)}{r_e^2} \quad (7.2)$$

where $\rho \gg 1$ and $(r - r_e) \ll r_e$. Here l is the free parameter given by $l = \frac{t}{\tau}$ where t is the time for light to travel the distance d between two black hole structures and τ is the time for light to travel through the whole throat region. The range $-\rho_c < \rho < \rho_c$ correspond to inner spacetime region of throat. Beyond this limit, the spacetime will be described by outer magnetic black hole spacetime discussed

previously. Author have discussed various reasons for the possibility of existence of this model of the wormhole in [maldacena2021humanly]. Their study involves discussion not only within classical general relativity but within all the known laws of physics. However, since this thesis is dedicated to understanding of observable properties of compact objects, we will discuss null geodesics, timelike geodesics and possibility of determining shadow shape of this wormhole in next sections.

7.2 Effective Potential for outer magnetic black hole spacetime

In this section we define the effective potential for the wormhole corresponding to null geodesics and timelike geodesics. However since this wormhole is described by two spacetime metrics, namely outer magnetic black hole spacetime and inner throat region spacetime, we derive expression of effective potential energy for both spacetimes separately. In this section, we discuss the effective potential of null geodesics and timelike geodesics for outer magnetic black hole spacetime.

Effective potential of null geodesics for outer magnetic black hole spacetime

From the discussion in previous chapters, the method of derivation of effective potential is well understood. For the spacetime of the outer magnetic black hole spacetime, the expression for total energy of null geodesics can be given using killing vectors and conserved quantities of energy and angular momentum as,

$$\frac{e^2}{l^2} = \frac{1}{l^2} \left(\frac{dr}{d\lambda} \right)^2 + \frac{1}{r^2} \left(1 - \frac{r_e}{r} \right)^2, \quad (7.3)$$

From the above expression, we can derive the expression of the effective potential energy of null geodesics corresponding to outer region of spacetime,

$$V_{eff} = \frac{1}{r^2} \left(1 - \frac{r_e}{r}\right)^2 \quad (7.4)$$

Fig.7.1 represents the above expression with effective potential energy of null geodesics as a function of the radial coordinate r . Differentiating the effective potential energy in the above equation 8.4 with respect to the radial coordinate r , and solving the expression for r , we get $r = 2r_e$ and $r = r_e$ for $\frac{d(V_{eff})}{dr} = 0$. Now using these two solutions of radial distance coordinate r in eq.8.4, we get two different values of effective potential energy at corresponding radial distances.

$$V_{eff}(2r_e) = \frac{1}{4r_e^2} \quad (7.5)$$

This corresponds to maximum value of effective potential energy of the null geodesics. While considering $r = r_e$ in eq.8.4, we get the following result,

$$V_{eff}(r_e) = 0 \quad (7.6)$$

which is minimum value of effective potential of null geodesics in outer magnetic black hole spacetime. These extremum values will correspond to unstable and stable circular orbits of light rays which leads formation of photonsphere around the spacetime.

Effective potential for a timelike case for outer spacetime

One can define conserved energy per unit mass and conserved angular momentum per unit mass for massive particles as discussed previously. Using these conserved quantities and normalization condition, the effective potential of massive particles for outer magnetic black hole spacetime can be given by,

$$V_{eff} = \frac{h^2}{2} \left(\frac{1}{r^2} + \frac{r_e^2}{r^4} - \frac{2r_e}{r^3} \right) \quad (7.7)$$

The graph of effective potential of massive particles in outer magnetic black hole

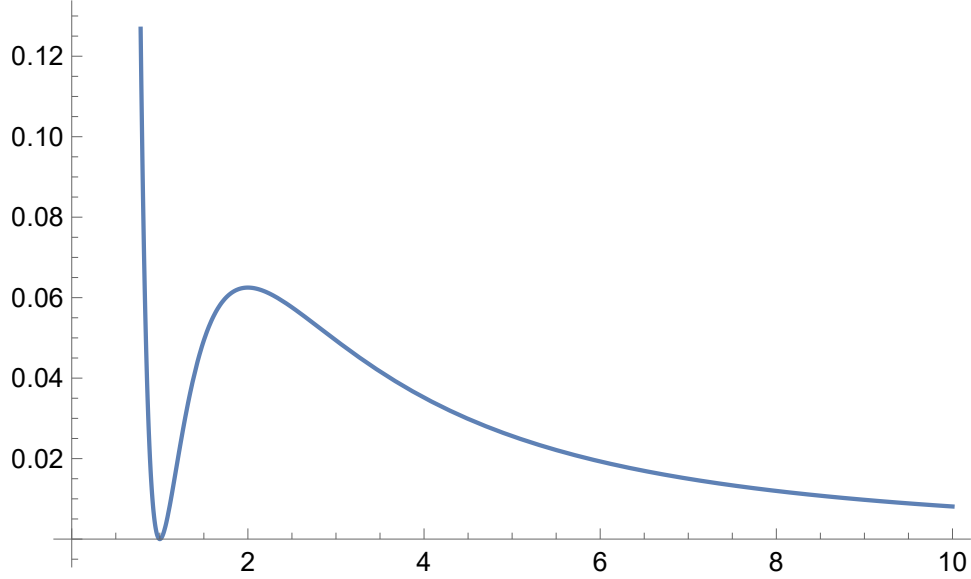


Figure 7.1: Effective Potential Energy plot for outer spacetime in nulllike case

spacetime with respect to the radial coordinate can be seen in 7.2. The horizontal line in the diagram corresponds to total energy. It can be seen that the line corresponding to effective potential energy intersects the horizontal line of total energy at two different values of radial distance. This suggests that massive particles moving on geodesic in the given outer magnetic black hole spacetime will have bound orbits. This will be discussed in the next part when the derivation of orbit equation of massive particle is discussed.

7.3 Effective Potential for inner throat spacetime

In this section, we discuss the effective potential of null geodesics and timelike geodesics for the inner spacetime metric which represents the throat region of the wormhole structure.

Effective potential for null like the case for throat spacetime

As discussed in the previous section, by defining the conserved energy per photon and conserved angular momentum per photon corresponding to the metric components of inner spacetime, the expression of effective potential energy of null geodesics can be written as,

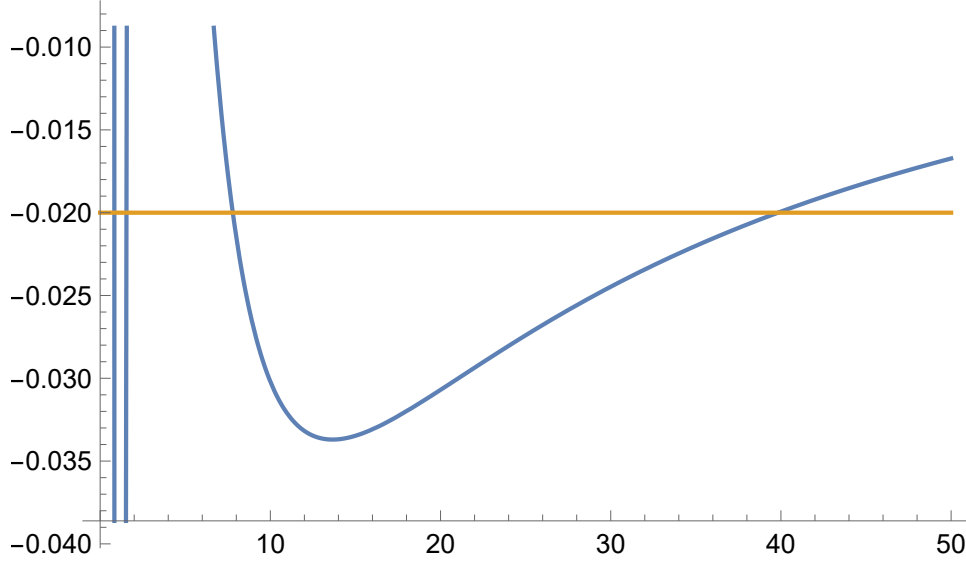


Figure 7.2: Effective Potential Energy plot for outer spacetime in timelike case

$$V_{eff} = (\rho^2 + 1) \quad (7.8)$$

The plot of effective potential energy of null geodesics with respect to the radial distance coordinate r can be seen in Fig.7.3. Here in the above eq.7.8, $\rho = l \frac{(r-r_e)}{r_e^2}$. However, since the range of is between $-\rho_c < \rho < \rho_c$, the above plot in Fig.7.3 can be understood as expanding in the opposite direction corresponding to negative values of the metric parameter ρ . The line corresponding to effective potential energy in the plot attains a finite value at $r = 0$ i.e. at the center of the throat length. Thus it suggests that photon will reach at that point and escape to the other side of the throat. It can also be observed that the potential attains minimum value at the radial distance $r = 1$, which is the value of the radial distance where the graph of effective potential energy of null geodesics for outer magnetic black hole region has the minimum value.

Effective potential for the timelike case for throat spacetime

As discussed earlier, one can define conserved energy per unit mass and conserved angular momentum per unit mass for massive particles. Using these conserved quantities and normalization condition, the effective potential of massive particles

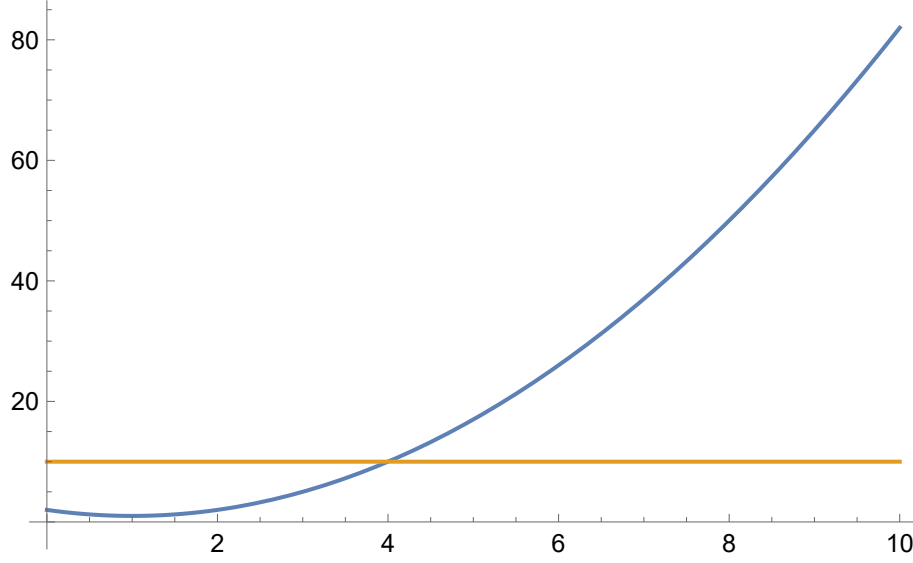


Figure 7.3: Effective Potential Energy plot for throat spacetime in nulllike case

for inner spacetime metric corresponding to throat region can be given by,

$$V_{eff} = ((\rho^2 + 1)(r_e^2 + h^2) - 1) \quad (7.9)$$

The graph of which can be seen in 7.4. Here also, there line corresponding to the effective potential energy is intersects the line corresponding to total energy of the timelike particles, thus one might think that massive particles will have bound orbits inside the throat region. However, there is a limit on the value of ρ . From which it can be noted that one can should not consider the graph upto $r = 1$ only since after that radial distance, the effective potential of the particle will be defined by effective potential of the outer magnetic black hole spacetime.

Here it is very important to that, since the wormhole is defined using two different spacetimes, the effective potential energy for null geodesics and timelike geodesics should be thought by considering the appropriate limits on the radial distance coordinates. For $r > 1$, the effective potential should be accounted only of outer magnetic black hole spacetime for both null and massive particles. While for $r < 1$, the effective potential should be accounted only of inner spacetime of the throat region for both null and massive particles. Interpreting the behavior of null and

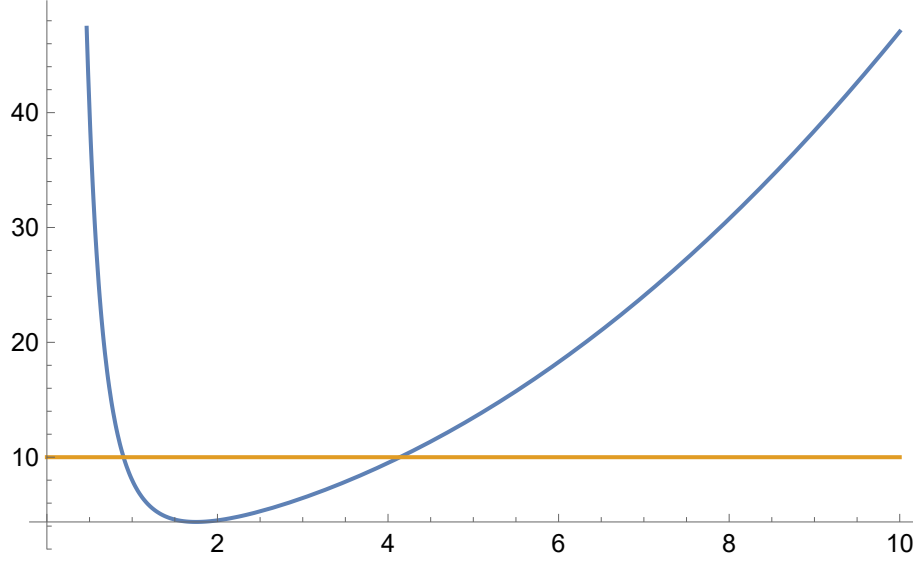


Figure 7.4: Effective Potential Energy plot for throat spacetime in timelike case

massive particle from the graph of effective potential energy outside the given limits would make no sense since both spacetimes are glued at the radius $r = 1$. To understand the motion of null geodesics and timelike geodesics perfectly, one needs to check the junction condition, which we discuss later in this chapter.

7.4 Orbit equations and orbit plot

7.4.1 Null Geodesics in Outer Spacetime

Considering an equatorial plane choosing polar coordinates $x = r \cos \phi$ and $y = r \sin \phi$, and assuming that light rays are coming from some far away light source and the line of rays is d distance away from the outer region spacetime structure, we can derive the orbit equation by first defining how ϕ is changing with r . Thus,

$$\frac{d\phi}{dr} = \frac{d\phi}{d\lambda} \frac{d\lambda}{dr} \quad (7.10)$$

where, $\frac{d\phi}{d\lambda}$ can be defined using angular momentum per unit rest mass as $\frac{d\phi}{d\lambda} = \frac{l}{r^2}$ and $\frac{d\lambda}{dr}$ can be written from the expression of total energy we get while deriving the equation of effective potential energy as,

$$\frac{d\phi}{dr} = \frac{l}{r^2} \frac{1}{l} \left(\frac{1}{b^2} - \frac{1}{r^2} \left(1 - \frac{r_e}{r} \right)^2 \right)^{-\frac{1}{2}} \quad (7.11)$$

now considering $u = \frac{1}{r}$, the above equation can be written as,

$$\frac{du}{d\phi} = \left(\frac{1}{b^2} - \frac{1}{r^2} \left(1 - \frac{r_e}{r} \right)^2 \right)^{-\frac{1}{2}} \quad (7.12)$$

from which, we can get the orbit equation,

$$\frac{d^2u}{d\phi^2} + u \left(1 - \frac{u}{u_e} \right)^2 - \frac{u^2}{u_e} \left(1 - \frac{u}{u_e} \right) = 0 \quad (7.13)$$

Plotting this equation in the Mathematica software, we get the orbits as can be seen in the plot 7.5. The graph represents that all the null geodesics approaching the wormhole with value of the impact parameter smaller than this limit will plunge into the wormhole and enter the throat region. While all the light rays approaching the wormhole with value of impact parameter higher than the above case will get deflected away without entering the throat region. These deflecting rays will contribute to the bright higher intensity region of shadow of this wormhole while the plunging light rays will contribute differently. In case of compact objects with singularity such as black hole and naked singularity, these plunging rays corresponds to the dark central region. But since we are here defining shadow for the wormholes, it is possible that light rays from plunging into other mouth of the throat might escape from this side of the wormhole and this might change the intensity plot of the shadow structure differently from the wormhole. As discussed earlier, to see this whole effect, we need to define the junction condition which we discuss after section 7.5.

7.4.2 Timelike Geodesics in Outer Spacetime

The derivation of the orbit equation of timelike particles for any general spherically symmetric static spacetime has been discussed in the chapter 5. So here we directly

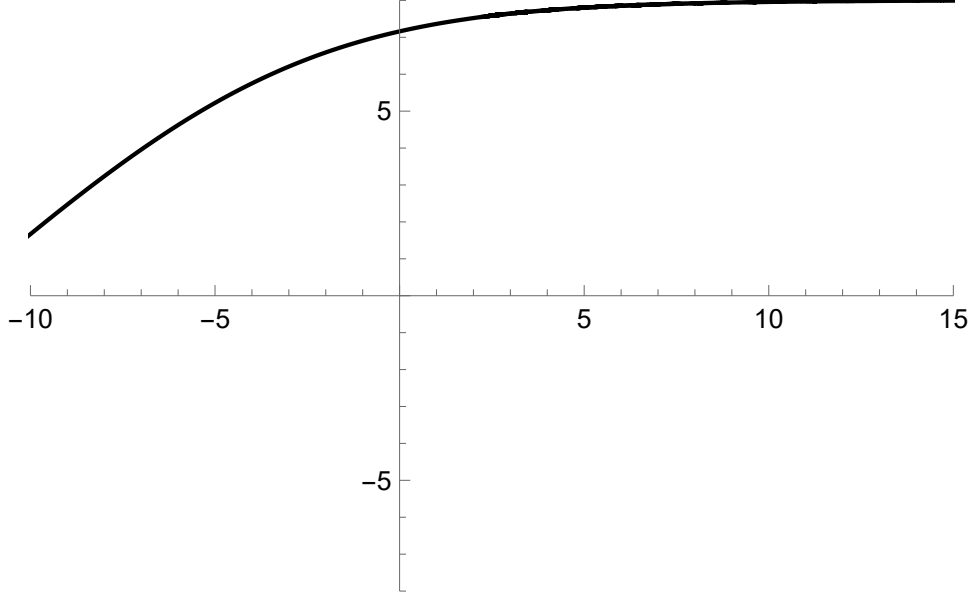


Figure 7.5: Null Geodesics in Outer Spacetime

use the general expression in eq.5.10 by putting the corresponding metric components of the outer magnetic black hole spacetime, from which we get the following expression,

$$\frac{d^2u}{d\phi^2} + u \left(1 - \frac{u}{u_e}\right)^2 - \left(\frac{u^2}{2} \left(1 - \frac{u}{u_e}\right)\right)^4 + \frac{1}{2h^2} \left(1 - \frac{u}{u_e}\right)^2 - \frac{\gamma^2}{2h^2} \frac{d\left(\left(1 - \frac{u}{u_e}\right)^2\right)}{d\phi} + \frac{\gamma^2}{2h^2} \left(1 - \frac{u}{u_e}\right)^2 \frac{d\left(\left(1 - \frac{u}{u_e}\right)^2\right)}{d\phi} = 0$$

As discussed earlier in chapter 5, it is very hard to obtain the analytic solution to the above expression since it is a second order differential equation. Thus we solve it numerically for corresponding values of conserved quantities of energy and angular momentum. Plotting this orbit equation in the Mathematica software, we get the orbits shown in the plot 7.6. From the graph, it is visible that the massive particles moves in bound orbits which what we had predicted while discussing about effective potential energy of massive particle for outer magnetic black hole spacetime. Also, it can be noted that there is positive precession of timelike orbits. This suggests if there exists a this wormhole spacetime, then motion of the stars moving around one



images/mbhtlfinal.pdf

Figure 7.6: Timelike orbits for spacetime of outer region

mouth of the wormhole will follow the precession.

7.4.3 Null Orbits for Throat Spacetime

To determine the trajectories of photons in the throat region of the wormhole, we need to again derive the orbit equation. For this purpose, we follow the same method as discussed previously in section 7.4.1. Thus we get the following expression,

$$\frac{d\phi}{dr} = \frac{d\phi}{d\lambda} \frac{d\lambda}{dr} \quad (7.14)$$

$$\frac{d\phi}{dr} = \frac{l}{r^2} \frac{1}{l} \left(\frac{1}{b^2} - (\rho^2 + 1) \right)^{-\frac{1}{2}} \quad (7.15)$$

From the above equation, we can obtain the orbit equation for null geodesics for the throat region as,



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Figure 7.7: Nulllike orbits for spacetime of throat region

$$\frac{d^2u}{d\phi^2} - \frac{2u_e^3}{u^2} \left(\frac{1}{b^2} - 1 \right) - \left(u^2 - \frac{u^3}{u_e} \right) \left(2u - \frac{3u^2}{u_e} \right) = 0 \quad (7.16)$$

As discussed earlier, it is hard to solve the above expression analytically. So we do it numerically for corresponding values of the conserved quantities of energy and angular momentum. The plot corresponding to the above orbit equation can be seen in the image 7.7. However, it is little bit hard to predict the motion of light rays only within the region $-\rho_c < \rho < \rho_c$ as the orbit equation takes all the values of radial distance coordinate r into account from 0 to asymptotic limit and it cannot consider the values of r corresponding to negative values of ρ which represents the other half part of the throat of the wormhole.

7.4.4 Timelike Orbits for throat region spacetime

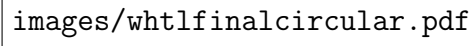
By using the general expression of orbit equation for a general spherically symmetric static spacetime derived in eq.5.10, we can define the orbit equation for massive particles in the inner throat spacetime of the wormhole as,

$$\begin{aligned}
& \frac{du^2}{d\phi^2} + \frac{u}{\frac{1}{u_e^2} \left(\left(u_e^2 \left(\frac{1}{u} - \frac{1}{u_e} \right) \right)^2 + 1 \right)^{-1}} \\
& - \left(\frac{u^2}{\frac{2}{u_e^4} \left(\left(\left(u_e^2 \left(\frac{1}{u} - \frac{1}{u_e} \right) \right)^2 + 1 \right)^{-1} \right)^2} + \frac{1}{\frac{2h^2}{u_e^4} \left(\left(\left(u_e^2 \left(\frac{1}{u} - \frac{1}{u_e} \right) \right)^2 + 1 \right)^{-1} \right)^2} \right) \\
& - \frac{\gamma^2}{2h^2 \left(\frac{1}{u_e^4} \left(\left(u_e^2 \left(\frac{1}{u} - \frac{1}{u_e} \right) \right)^2 + 1 \right) \right) \left(\frac{1}{u_e^4} \left(\left(u_e^2 \left(\frac{1}{u} - \frac{1}{u_e} \right) \right)^2 + 1 \right)^{-1} \right)^2} \\
& \quad \frac{d \left(\frac{1}{u_e^2} \left(\left(u_e^2 \left(\frac{1}{u} - \frac{1}{u_e} \right) \right)^2 + 1 \right)^{-1} \right)}{du} \\
& + \frac{\gamma^2}{2h^2 \left(\frac{1}{u_e^4} \left(\left(u_e^2 \left(\frac{1}{u} - \frac{1}{u_e} \right) \right)^2 + 1 \right) \right)^2 \left(\frac{1}{u_e^4} \left(\left(u_e^2 \left(\frac{1}{u} - \frac{1}{u_e} \right) \right)^2 + 1 \right)^{-1} \right)} \\
& \quad \frac{d \left(\frac{1}{u_e^2} \left(\left(u_e^2 \left(\frac{1}{u} - \frac{1}{u_e} \right) \right)^2 + 1 \right) \right)}{du} = 0
\end{aligned}$$

As discussed earlier, it is hard to solve the above expression analytically. So we do it numerically for corresponding values of the conserved quantities of energy and angular momentum. The plot corresponding to the above orbit equation can be seen [7.8](#) and in [7.9](#). These two plots corresponds to different values of parameters. However, as mentioned in the previous section, it is hard to interpet the motion of light rays only within the region $-\rho_c < \rho < \rho_c$.

7.5 Junction Conditions

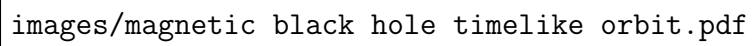
As discussed previously, the wormhole model proposed by authors in [[maldacena2021humanly](#)] is represented by two different spacetime metrics. The outer region of the spacetime



images/whtlfinalcircular.pdf

[b]

Figure 7.8: Timelike circular orbits for spacetime of throat region



images/magnetic black hole timelike orbit.pdf

Figure 7.9: Timelike bound orbits for spacetime of throat region

is represented by magnetic black hole spacetime, while the inner throat region is represented by a different spacetime metric. Since the whole spacetime geometry consists of two different spacetimes, it is important to check if the junction conditions are being satisfied by these spacetimes. Also we have the equations of orbits for timelike as well as nulllike particles for both of these spacetime metric. However, to determine the complete trajectories of any particle moving in the outer region of the spacetime, coming near the mouth of the wormhole and passing through the throat, one needs to have the spacetime metric representing the geometry of the region connecting the outer region and the throat region. The metric in this region can be defined by using junction conditions. The junction conditions are:

[1] The induced metrics of internal and external spacetimes on the matching hypersurface should be identical with each other.

[2] Another condition is to match the extrinsic curvatures (K_{ab}) at the hypersurface. Extrinsic curvature can be expressed in terms of the covariant derivative of normal vectors on the hypersurface: $(K_{ab}) = e_a^\alpha e_b^\beta \nabla_\alpha \eta_\beta$, where e_a^α is the tangent vectors on the hypersurface and η_β is the normal to that hypersurface.

Thus following the above mentioned conditions, we can define hypersurface at the radius r_e where two spacetimes will be matched together with each other. However, from our investigation, it was found that the junction conditions are not perfectly satisfied for the above mentioned spacetime metrics. Because of this reason, we cannot determine complete trajectories of null geodesics and thus shadow cannot be defined perfectly.

Chapter 8

Gravitational Lensing

Universe consists countless planets, stars, star clusters, galaxies, compact objects such as a black hole at the center of the M87 galaxy, etc. These structures are basically matter distribution. And thus, from Einstein field equations, we know that it introduces curvature in the spacetime continuum. Thus light coming from any source such as star etc would follow the spacetime curvature and gets bent. When huge mass, because of its gravitational field, deflects and magnifies the light coming from far away galaxies behind the mass itself, then this phenomenon is known as gravitational lensing. Compact objects are objects with huge mass in its geometry so they can also produce gravitational lensing. In this chapter, we are going to study gravitational lensing by compact objects.

8.1 Gravitational lensing of light around Schwarzschild Black Hole

Consider a Schwarzschild black hole in the spacetime and light ray coming from any far away source near the Schwarzschild black hole, gets deflected and escapes to infinity as it is not plunging into the black hole. For such scenario, deflection of light can be defined by change in azimuthal coordinate $\Delta\phi$. We can derive this $\Delta\phi$ from the equation 2222 integrating the expression with respect to r ,

$$\frac{d\phi}{dr} = \frac{l}{r^2} \frac{1}{l} \left(\frac{1}{b^2} - \frac{1}{r^2} \left(1 - \frac{2M}{r} \right)^2 \right)^{-\frac{1}{2}} \quad (8.1)$$

integrating above equation with respect to the radial distance coordinate r , we get,

$$\Delta\phi = \int_{r_1}^{\infty} \frac{1}{r^2} \left(\frac{1}{b^2} - \frac{1}{r^2} \left(1 - \frac{2M}{r} \right)^2 \right)^{-\frac{1}{2}} dr \quad (8.2)$$

here $r_1 = r$ for which $V_{eff} = \frac{1}{b^2}$.

To study how deflection angle changes with respect to the impact parameter b , one can introduce a new factor ω which can be given by $\omega = \frac{b}{r}$. Thus equation 2222 becomes,

$$\Delta\phi = 2 \int_0^{\omega_1} \left(\frac{1}{b^2} - \omega^2 \left(1 - \frac{2M\omega}{b} \right)^2 \right)^{-\frac{1}{2}} d\omega \quad (8.3)$$

here ω_1 is the value of ω for which the expression of $\Delta\phi$ depends only on the $\frac{M}{b}$. Integrating this expression of $\Delta\phi$ in terms ω , we get the final expression,

$$\Delta\phi = 2 \left[(M^2 + b^2) \arcsin \left(\frac{b\omega - M}{(M^2 + b^2)^{\frac{1}{2}}} \right) - \frac{((b)(b - \omega)(b\omega - 2M))^{\frac{1}{2}}}{b^2} \right]_0^{\omega_1} \quad (8.4)$$

and putting values of limits and writing the expression for small approximations,

$$\Delta\phi = \pi + \frac{4M}{b} \quad (8.5)$$

Now from figure given, it can be understood that expression for deflection angle $\delta\phi$ can be written in the form of $\Delta\phi$ as,

$$\delta\phi = \Delta\phi - \pi \quad (8.6)$$

Thus, from equation 22 and 33, we can write deflection angle $\delta\phi$,

$$\delta\phi = \frac{4M}{b} \tag{8.7}$$

8.2 Strong Deflection Angle

However, there is another method to define the deflection angle, known as strong deflection limit. The strong deflection limit suggests the scenario where photons, after reaching near the black hole, rotates around it and then escapes to the infinity. Observer in the asymptotic region can observe the photon that revolved for $\frac{\pi}{2}$ angle in the shape of a ring, which will be known as primary ring. The observer might also observe the photons that has completed $\frac{3\pi}{2}$ or $\frac{5\pi}{2}$ rotating in the form of the ring which would be known as secondary ring and first higher order ring respectively.

For my MSc dissertation work, We have calculated the effective pottential for nulllike and timelike case for metric representing the out region around the wormhole and also for the metric of the deep throat region. We then derived orbit equations for from the expression of effective potential energy for timelike orbits and nulllike orbits for both of the metric. We tried finding path of any particle moving from outer region around the wormhole to the deep throat region. But the original paper titled "humanly traversable wormhole" by Prof. Juan Maldacena does not contain the metric representing the region connecting two spacetime geometries. We tried to derive the spacetime of that region using matching condition. But the metric components of both the spacetimes are in such a way that the equation blows up on matching the metric for one certain radius. And thus we cannot define the complete trajectory of the particle coming from outer region and passing tthrough the throat region, thus cannot also define how shadow would look like for this wormhole.

In future, we are going to work on this phenomenon as the higher order rings may give additional details about the metric and the spacetime geometry around the two compact objects and this technique might be helpful in distinguishing two different compact objects.

Chapter 9

Conclusions

9.1 Conclusions and Results

In this thesis work, we wanted to examine some properties of the compact objects which can possibly act as observational signatures to probe the nature of the compact objects at the galactic centre and also to distinguish different types of compact objects from each other. For this, we mainly focused on three types of compact objects: Black Holes, Naked Singularities, Wormholes. Black holes are objects with central singularity covered by event horizon around it. Formation of black holes can be understood by considering gravitational collapse of a matter cloud with homogeneous matter distribution and zero pressure components. However, as discussed in chapter 4, many studies suggest that when more realistic physical conditions were taken into consideration such as inhomogeneous matter cloud with non zero pressure components, the structure of an end state does not possess horizons as a result of gravitational collapse depending upon the equilibrium conditions. This suggests that naked singularity might also form under certain scenario. Other than this, we considered another type of compact objects which are exact solutions to the Einstein's field equations, namely wormhole and have been a topic of study because of interesting properties.

We studied shadow and timelike orbits for the above mentioned compact objects.

The reason behind studying the shadow properties is that Event Horizon Telescope is observing the compact objects at the galactic centres. In 2019, they released image of a compact object at the centre of M87 galaxy. Also in May, 2022, they will be releasing the image of compact object at the centre of our galaxy, Milky Way. Thus this property is an important aspect since theoretically generated shadow images can be used to probe the nature of these compact objects. While the reason behind studying the timelike orbits is, there are many stars which are revolving around the central compact object of our galaxy. The motion of these stars can be studied as to determine the nature of the compact object. This is because the timelike orbits depend on the metric components of the compact objects spacetimes.

In chapter 3, we discussed about Schwarzschild black hole, general axisymmetric rotating compact objects and Kerr black holes. We also discussed the concept of shadow, derivation of effective potential energy for any general spacetime, null geodesics in spherically symmetric static spacetimes, shadow of Schwarzschild black hole. We also discussed the observable intensity of shadow in one section. Later we discussed details about shadow and photonspheres for Schwarzschild spacetime. For rotating compact objects, we discussed the general formalism to understand the motion of null geodesics and a brief discussion about effect of rotation on the shadow shape was also included.

In chapter 4, we first discussed about formation of naked singularities from gravitational collapse very briefly. Later we studied the properties of JMN-1 naked singularity and JNW naked singularity spacetimes. For these two spacetimes, we discussed about effective potential energy for null particles corresponding to these spacetimes from which shadow shape and photonsphere can be determined.

In chapter 5, we discussed the derivation of orbit equations for timelike geodesic orbits in a general spherically symmetric and static spacetime. Later we derive the

orbit equation for Schwarzschild spacetime where information regarding Innermost Stable Circular Orbits ISCO is also included. In the later sections of this chapter, we define the orbit equations of timelike particles in JMN-1 and JNW spacetime metric.

In chapter 6, we give literature review regarding wormholes. We first define the expansion scalar for a wormhole metric to get an idea about geometrical shape of the spacetime. Later we discuss physical properties associated with wormhole spacetimes. In particular, we Ellis wormhole and Morris-Thorne class of wormholes and various details about these spacetimes such as flare out conditions, redshift function and shape function. The stability of throat of the wormhole is satisfied by violating the energy condition within classical general relativity, which is discussed using focusing theorem. Later we define the effective potential of the Morris-Thorne wormholes. A detailed discussion regarding shadow shape etc for this wormhole has not been included as it is very well defined in the literature.

In chapter 7, we discuss about humanly traversable wormhole proposed in [maldacena2021humanly]. We discuss some details regarding this proposed wormhole spacetime. The spacetime geometry is defined by two different spacetime metrics. Later we define the effective potential energy for null and massive particles corresponding to both spacetime metric and plot the graph of effective potential energy with respect to the radial distance coordinate. We later derive the orbit equations for timelike particles and null particles for both, inner throat spacetime and outer magnetic black hole spacetime. We also discuss some features of these observational properties. In the last section, we discuss about junction conditions for matching of spacetime metrics at a hypersurface.

As a learning outcome, we understood shadow property and timelike geodesics can be very useful features to know details about the compact object spacetimes observationally. The formation of photonsphere and observable intensity can be used

to distinguish compact objects from each other. Apart from this, the information regarding precession of timelike orbits can also be useful for the same purpose of determining nature of the compact objects. Apart from this, we also learnt some aspect of gravitational collapse of homogeneous and inhomogeneous matter cloud which results into the formation of black holes and naked singularity depending upon the equilibrium condition achieved. We also understood that existence of wormholes requires violation of energy conditions only within classical general relativity. While it can be perfectly explained using quantum theory within the laws of physics.

For the humanly traversable wormholes as proposed by authors in [**maldacena2021humanly**], we investigate that there can be existence of photon sphere in around one mouth of the wormhole in the outer magnetic black hole spacetime. Also we determine that timelike particles can be in bound orbits in this spacetime and their orbits follow the positive precession which is also observed in the Schwarzschild spacetime. It is also determined that photon coming from asymptotic infinity can enter in the throat of the wormhole and pass through it to escape from the other side of the wormhole structure. The motion of timelike particles in the throat region cannot be defined well. This is because the junction conditions which should be fulfilled for the matching of two spacetimes are not defined particularly. We are still investigating the possibility to define the spacetime at hypersurface so that we can define the shadow for this wormhole spacetime.